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Meet Team Tri-Side Die



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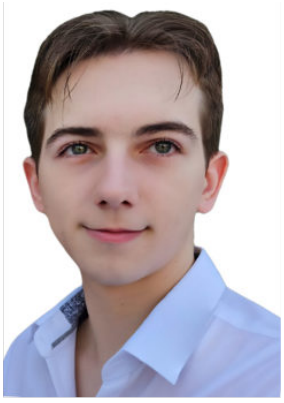
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1. Design History

Changelog

When	What	Who
14/03/2026	2 - Vision statement	eduardionut.zugravu@st…
20/03/2026	3.1 - Target audience	eduardionut.zugravu@st…
21/03/2026	3.2 - Platform	eduardionut.zugravu@st…
30/03/2026	3.3 - System requirements	eduardionut.zugravu@st…
02/04/2026	3.1 - Target audience, Refined	Marco Caramazza
02/04/2026	4 - Legal analysis	Marco Caramazza
03/04/2026	3.4 - Top performers	eduardionut.zugravu@st…
04/04/2026	5.2 - Gameplay description, Hive Storing, save and load	Daniel Bonardi
04/04/2026	5.2 - Gameplay description, Cell Editing	Daniel Bonardi
04/04/2026	5.1 - Gameplay overview	eduardionut.zugravu@st…
04/04/2026	5.2 - Gameplay description, Hive Storing, save and load	Daniel Bonardi
04/04/2026	5.2 - Gameplay description, Cell Editing	Daniel Bonardi
05/04/2026	5.2 - Gameplay description, Time and Events	Marco Caramazza
06/04/2026	5.3 - Controls, Bindings	Daniel Bonardi
06/04/2026	5.2 - Gameplay description, Breeding	Marco Caramazza
08/04/2026	5.2 - Gameplay description, Resources	eduardionut.zugravu@st…
08/04/2026	6 - Game characters	Marco Caramazza
09/04/2026	7 - Game world	Marco Caramazza
10/04/2026	5.3 - Controls, Procedures	Daniel Bonardi
10/04/2026	5.3 - Controls, Interfaces	Marco Caramazza
11/04/2026	5.3 - Controls, Rules	eduardionut.zugravu@st…
11/04/2026	5.3 - Controls, Behaviors	eduardionut.zugravu@st…
12/04/2026	5.4 - Modes	eduardionut.zugravu@st…
13/04/2026	7 - Game world, added weather events	Daniel Bonardi

Changelog

When	What	Who
13/04/2026	3.5 - Feature comparison	eduardionut.zugravu@st…
15/04/2026	3.6 - Personas	eduardionut.zugravu@st…
15/04/2026	5.5 - Flowchart	Daniel Bonardi
15/04/2026	Added cover page	Marco Caramazza
16/04/2026	8 - Media list	Marco Caramazza
16/04/2026	9 - Prototype summary	Daniel Bonardi
01/07/2026	Added new competitor (Sim Ant)	Marco Caramazza

2. Vision Statement

Hivemind is an artificial life simulation, specifically a bee colony management game set in the modern world. Players guide a colony of bees through challenges and harsh conditions, striving not only to survive but to become the most powerful hive, as a wise queen once said: “Surviving is the goal of ordinary creatures; the true ambition is to become the most powerful hive of all.” To achieve this, players must carefully manage their resources to keep production running; after all, honey doesn’t make itself. Even the strongest bees are not immortal or omnipotent; they need care and attention. Monitoring their moods and customizing the hive through a furnishing system are essential to maintain a happy and efficient workforce.

Bees form the backbone of the colony, and players must send them foraging to gather nectar, pollen, resin, and water, the four primary resources critical for survival. These expeditions are not without risk, as random events may arise along the way, forcing players to make choices that can lead to different outcomes. At the same time, the hive faces external threats such as storms, hornet attacks, and changing climates, reminding the player that life outside the hive is as perilous as the work inside it. The colony’s fortress is made of hexagonal units forming various rooms that can be expanded, furnished, and equipped with production units. One vital room is the brood room, where the queen lays eggs and new bees are raised, ensuring the continuation and growth of the colony.

Managing resources extends beyond simple gathering. Nectar, pollen, resin, and water are used directly, but the hive can also craft essential goods: glue is produced by combining pollen and resin, food from nectar and water, and honey directly from nectar. Balancing collection and crafting is key to sustaining the colony and expanding its capabilities, allowing the player to prepare for both everyday survival and extraordinary events.

As the colony grows, players can experience a deeper layer of strategy by connecting with up to three other players, merging hives into a single collective. In this cooperative mode, resources are shared proportionally, and decisions must be coordinated, giving players the experience of acting as a true hivemind. This shared management encourages players to consider both the strengths and weaknesses of their combined bees, highlighting the importance of specialization and diversity within the hive.

Not all bees are the same, and specialization plays a crucial role. The colony includes different species of bees, each adept at specific tasks, from foraging to defending the hive. Through a breeding system, players can produce new bees and unlock entirely new species. This feature becomes especially exciting in multiplayer, where combining the bees of different players can yield unique species, granting the hive new abilities and opening up fresh strategic possibilities. In this way, the growth of the hive, the careful management of resources, and the cooperative merging of colonies all tie together into a single hivemind.

2.1 Game logline

Build, Expand, Evolve – and make some Honey!

2.2 Gameplay Synopsis

The gameplay experience is mostly situated inside the hive, where the player manages the bees from a lateral-cutaway perspective (as if a section has been sliced off, revealing the hexagonal rooms). Around the internal view of the hive there is the frame user interface, which allows the player to interact with the hive, overseeing the resources dynamics and info about its own bees. The player is also capable of issuing direct orders to the single bees, through the frame UI. On the top-left corner of the frame UI there is an hexagonal representation of the resources (one hexagon for each resource). On the top-center there’s the timeline, showing past events on the left and upcoming ones on the right. The present moment is centered and indicated by a play/pause button, which has other three buttons around to adjust the game speed (1x - 2x - 3x).

The game starts with a basic hive structure, composed of seven hexagonal rooms and a nursery. In terms of workforce, the player is provided initially only with the queen (essential for the colony continuity). Then the game asks the player if any tutorial is needed, after that the player can begin his adventure, with challenges and opposing events, building and expanding the hive, and steadily increasing the size of the colony day by day. Potentially a match can last forever, there isn’t an end to this adventure, only the player can decide when is the time to wrap up.

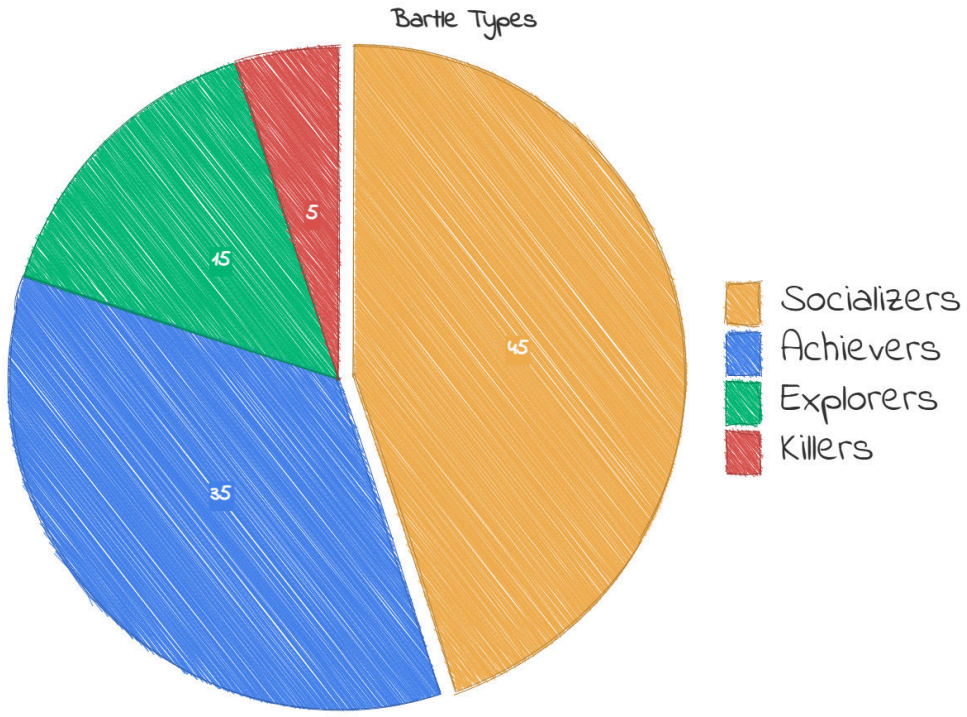
3. Audience, Platform, and Marketing

3.1 Target Audience

Hivemind appeals to players who enjoy resource management, problem-solving, cooperation, discovery, and a game that requires a significant time investment. The game evolves slowly, and allows players to write their own story over many hours of gameplay, while offering constantly new, unpredictable challenges that players have to adapt to. But it also embraces that slice of audience that prefers to lay back sometimes and enjoy the game play itself.

3.1.1 Bartel Types

The game aims to target the socializer types, but we believe the game can be quite appealing to achievers, who may try to find ways to exploit and fully optimize the gameplay, and explorers who wish to discover everything the game has to offer:



- **Killers (5%):**
  May enjoy some of the challenges the game has to offer, but the gameplay offers no opportunity to engage against other players; in fact, there are many safeguards that are put in place explicitly to make the game as safe as possible from griefers.
- **Explorers (15%)**:
  Will find great enjoyment in discovering all the bees, mutations, and events the game has to offer, but they might find the limitations of the hive unrewarding or quickly run out of the limited content without engaging with the more technical aspects of the game.
- **Achievers (35%):**
  The game offers players plenty of room to be mastered, even to the point of abusing some of the mechanics by deeply learning how each gameplay aspect functions. The game warmly welcomes players who excel at optimization and automation, creating hives that are entirely self-sustained.
- **Socializers (45%):**
  Our main target audience. At its core, Hivemind is a game meant to be played with friends to achieve its full potential. Everything about the game is geared towards a fun collaborative experience. The sharing of resources, the union of the hives, the collaborative building. While the game is perfectly playable in single-player, players will find much greater enjoyment from joining their hive with some friends and experimenting together.

3.1.2 Demographics

Age Distribution

From an analysis done on the recent playtime trend on games that fill the same niche as Hivemind, we were able to draw some important information on our target audience:

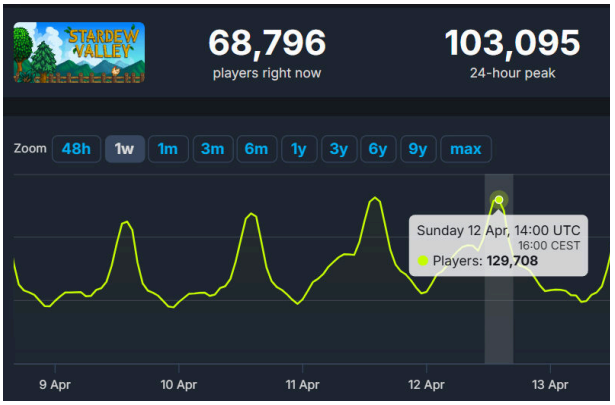




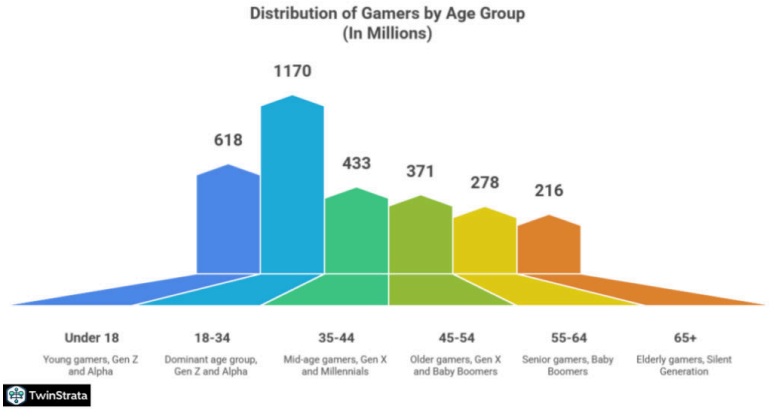
Taking into consideration that games like Spore were released 18 years ago, the trends that we are seeing represent the most dedicated player base of the game. Over a week, both **The Sims 4** and **Spore** reach their peaks at around 18:00 UTC. It is safe to assume the playerbase is most active **around work hours.**

We found these trends appropriate for our game due to the fact that the hardcore nature of Hivemind does not come from technical difficulties, but rather from a time investment perspective. But nonetheless, to be more certain of our analysis, we also looked at managerial games with much more technical depth, like **Shadow Empire**, and long-standing games with cooperative play known for their great time investment, like **Stardew Valley**, and found fairly similar trends.

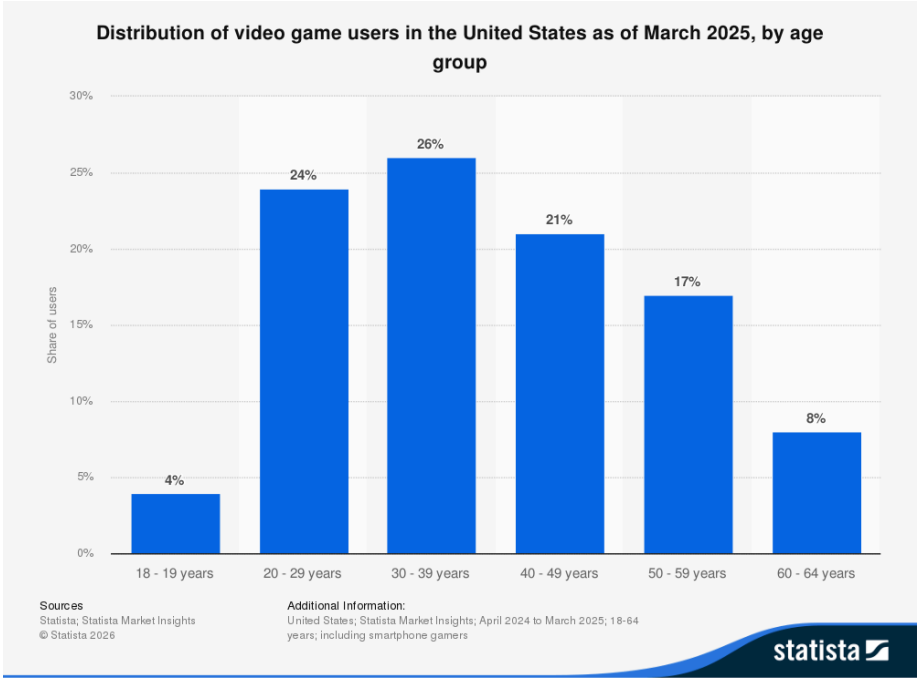




Finally, according to this report https://www.twinstrata.com/gamers-statistics/: youth under 20 mainly play 'disposable' games, not titles that require continuity and strategy. So we feel safe estimating our target age will be between 25-44.

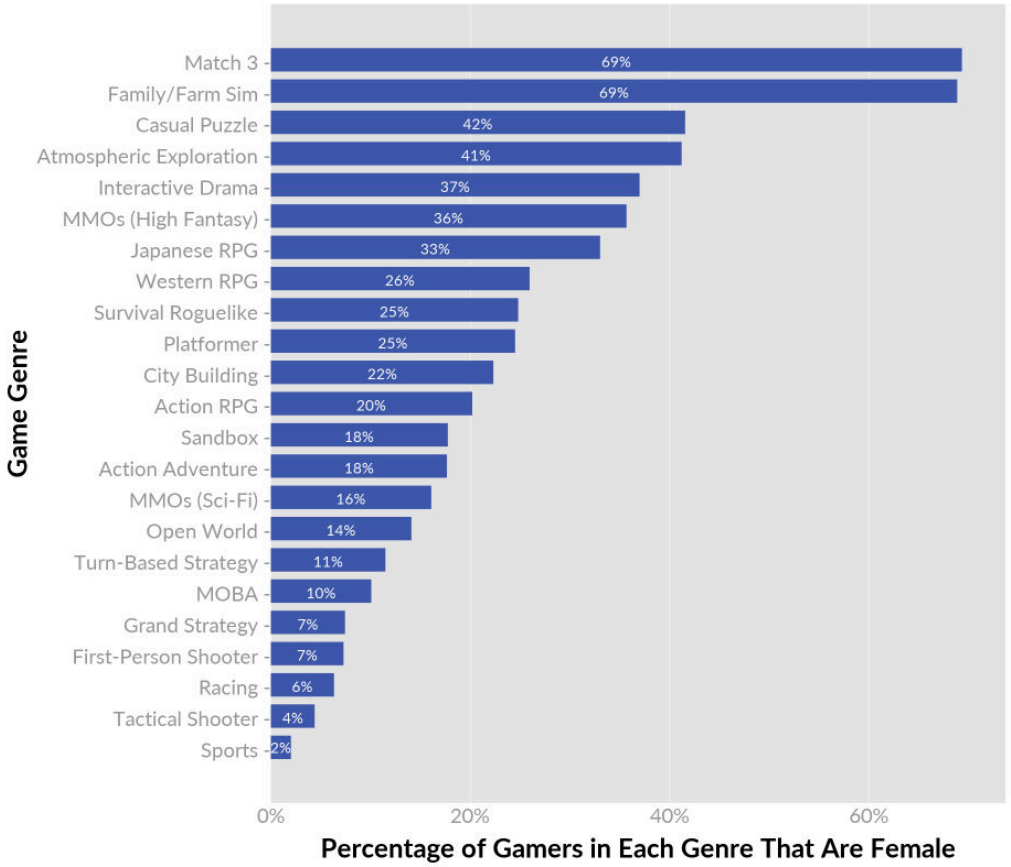


Which matches the distributions published by Statista in 2025 , showing that our game would embrace the two largest slices of video gamers in the US as of March 2025.



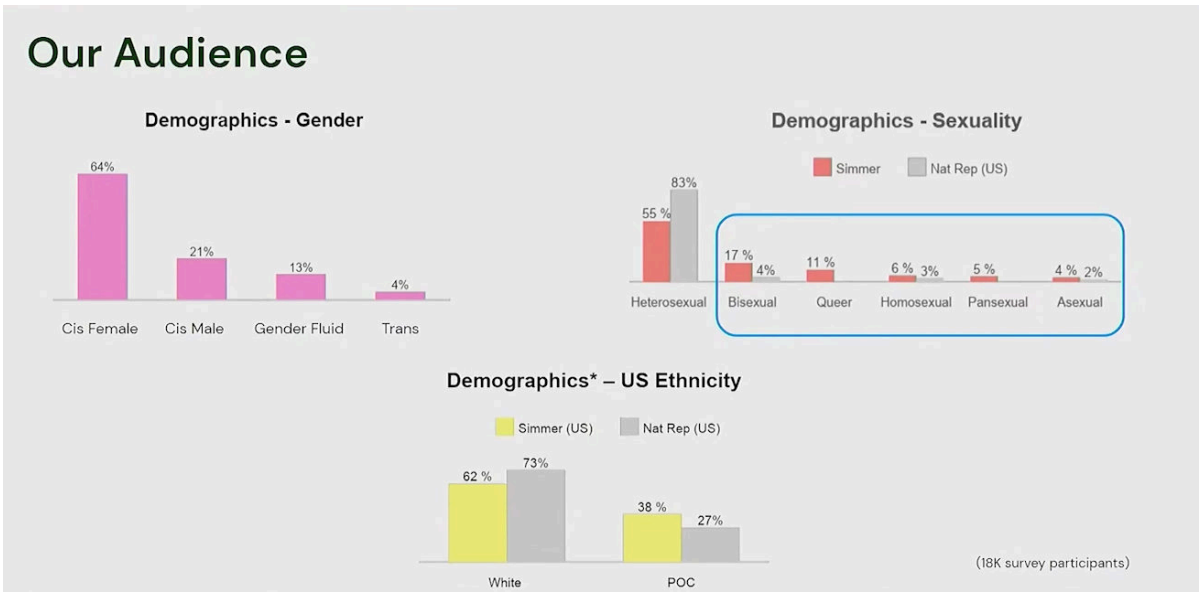
https://www.statista.com/forecasts/1277856/video-game-users-age-market-usa/

Gender Distribution
According to this blog dated 2017 breaking down the percentage of female gamers by genre, it is evident women prefer games that involve simulation, around 69% in fact, and also shows that a non-trivial 22% enjoys City Building games, which tend to involve management aspects.



https://quanticfoundry.com/2017/01/19/female-gamers-by-genre/

Philip Ring, Executive Producer on The Sims 4, shared some important insights on the game on GDC in 2023, specifically regarding their audience demographics:



https://www.gdcvault.com/play/1029364/-The-Sims-Process-and

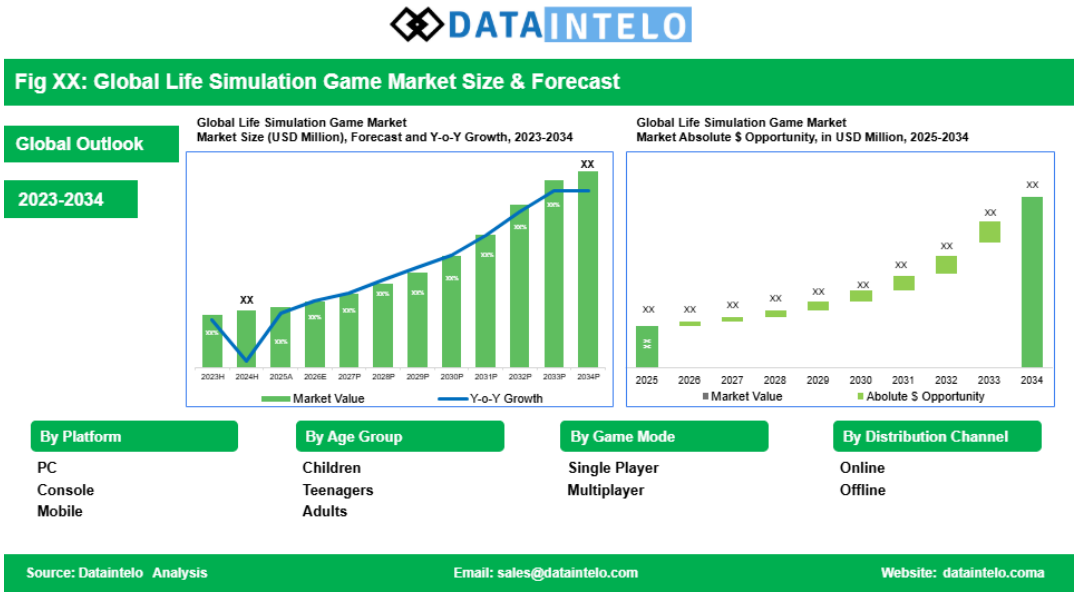
Showing a 64% of the player base identifying as Cis Females, only a 21% Cis Males, and the non-negligible remaining 17% Gender Fluid and Trans gamers.
Following these considerations, we expect a majority of our player base to be **Cis Famales**, but we kept our game entirely gender neutral to also embrace the portion of gamers within the LGBTQ+ community, which seems to gravitate more towards Simulation games with cozy art styles.

Geographic
Finally, while it's difficult to find a precise forecast for our specific niche, this report analyzing the growing trends of **Online Simulation Games** would suggest our biggest source of revenue would come from continents like **Asia** (notably with the exception of Russia) and **Africa**. It specifies: *'By geography, Asia-Pacific led with 44.70% of online simulation games market share in 2025, while the Middle East & Africa region is projected to expand at a 17.62% CAGR through 2031'.*



https://www.mordorintelligence.com/industry-reports/global-online-simulation-games-market

This second report, on **Life Simulation Games**, reaffirms **China, Japan** and **South Korea** as the major contributors to the market, emphasising China as a '*dominant force*' in the market, while also pointing out the significant market presented by **North-America** thanks to their '*high levels of disposable income and a strong presence of major gaming companies*', the latter notably including **Electronic Arts**:



https://dataintelo.com/report/life-simulation-game-market

3.2 Platform

The simulation genre is primarily associated with PCs. This is not a coincidence, as the mouse-and-keyboard input system allows players a high level of interaction with the complex mechanics of simulation and management games. The experience is often unsuitable for portable consoles, handhelds and mobile due to the large amount of information that needs to be displayed on-screen, and controller play would also be rendered tedious by the mechanical limitations of the peripheral.



Civilization VII
In terms of amount of visual information, and comfort with a controller, our game and Civ 7 are comparable

In terms of PCs, there are also different operating systems to manage. For this reason as an indie simulation game, the main OS target is Windows as everyone knows, it is the main platform for PC gamers.

3.3 System requirements

The following requirements are based on similar games in terms of assets and scale, like Stardew Valley, Rimworld, Factorio, Terraria and Mewgenics.

Minimum requirements:

CPU	Intel Pentium G4560 AMD Athlon 3000G
RAM	4 GB
GPU	Intel HD 4000 AMD Radeon R5 230 or equivalent integrated graphics
Storage	2 GB
OS	Windows 10/11 - 64 bit
DirectX	Version 11
Target resolution	720p @ 30fps
Connection	Download: 5 - 10 Mbps Upload: 3 - 6 Mbps
Additional notes	Input system with mouse and keyboard only

Recommended requirements:

CPU	Intel Core i3-10100 AMD Ryzen 3 3100
RAM	8 GB
GPU	Nvidia GTX 1050 AMD RX 560 or equivalent integrated card
Storage	3 GB
OS	Windows 10/11 - 64 bit
DirectX	Version 11
Target resolution	1080p @ 60fps
Connection	Download: 15 - 25 Mbps Upload: 10 - 15 Mbps
Additional notes	Input system with mouse and keyboard only

3.4 Top performers







As an honorable mention, we include **APICO** for its similarities with **Hivemind**



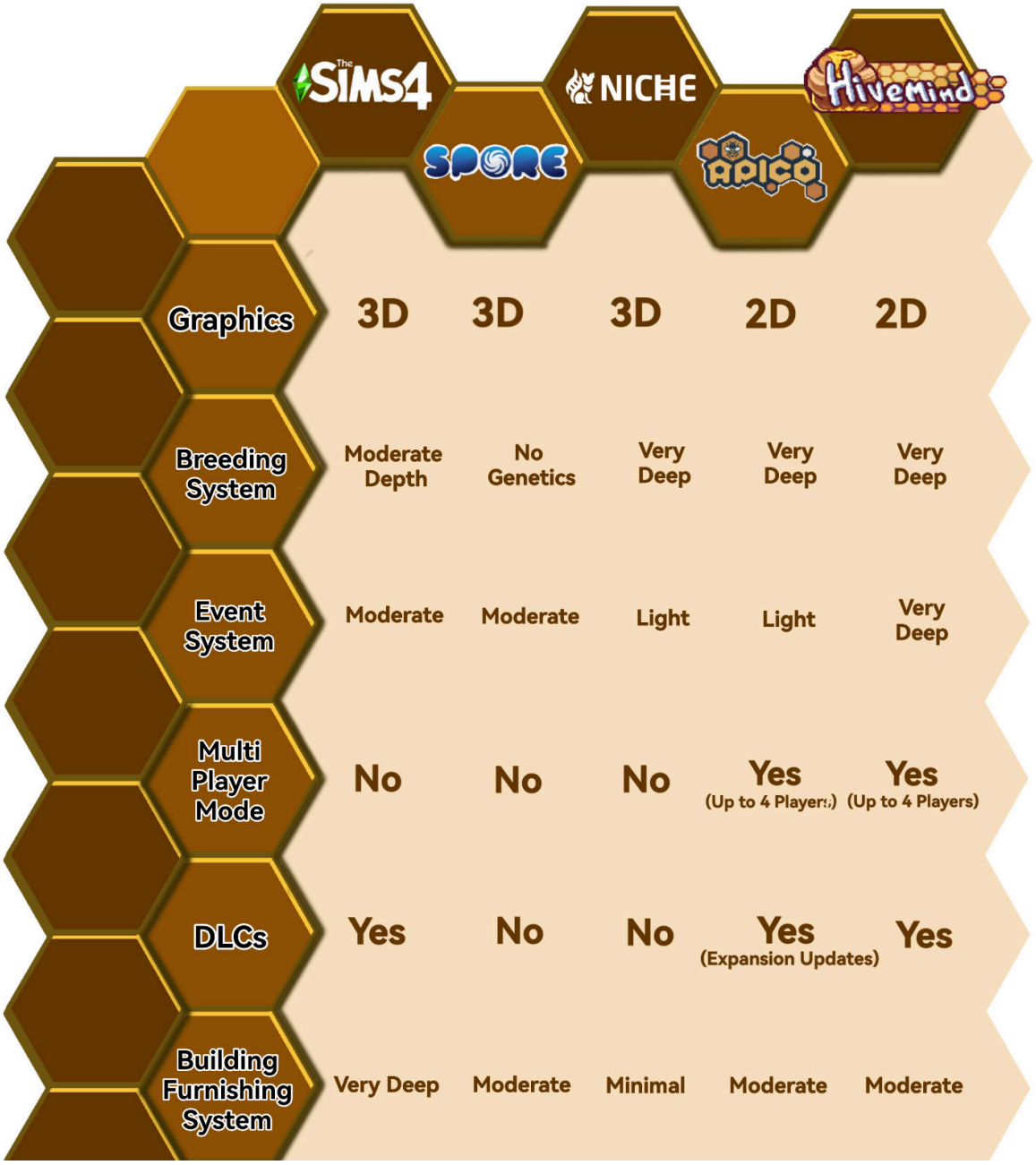
And **SIMANT**



3.5 Feature comparison



All* = Windows - Mac OS - Linux - Nintendo Switch - PlayStation 4 & 5 - Xbox One/S/X


The SIMS 4
NICHE
HiveMind
SPORE
apico
Graphics
3D
3D
3D
2D
2D
Breeding System
Moderate Depth
No Genetics
Very Deep
Very Deep
Very Deep
Event System
Moderate
Moderate
Light
Light
Very Deep
Multi Player Mode
No
No
No
Yes (Up to 4 Players)
Yes (Up to 4 Players)
DLCs
Yes
No
No
Yes (Expansion Updates)
Yes
Building Furnishing System
Very Deep
Moderate
Minimal
Moderate
Moderate

3.6 Personas


Personal info
Sindy Altman
From US - Detroit
Age: 41
Occupation
Sales assistant
Technology and Videogames Knowledge
Technology: 3/5
Videogame: 4/5
Goal
Sindy wants to find out every aspect of the game !!!
Background Description
Sindy Altman is a sales assistant who has worked in retail for over ten years, known for her reliability and customer skills. She started to play Videogames in her teens on a family PC and still plays regularly.
Bartle's Type
Socializer
Explorer
Criteria for Videogames
Relaxing and chill cooperation with my friends
Platforms
Windows
Nintendo Switch
Frustration with Videogames
She doesn't like competition, instead Sindy believes in hard work and cooperation.


Student
Occupation
Haruto Tanaka
Japan - Hosaka
Age: 26
Personal info
He is always looking for a match against the Computer
Goal
Technology: 4/5
Videogame: 6/5
Technology and Videogames Knowledge
Haruto studies economics at university. He is a remarkable student as he only needs to listen the lectures and to study a little to perform very well. These skills allow him to spend a lot of his free time on playing Videogames, hunting down every aspect.
Background Description
Haruto wants to face challenges and manage resources carefully
Criteria for Videogames
Achiever
Bartle's Type
He doesn't like to cooperate with other players, Haruto is a solo adventurer
Frustration with Videogames
Windows
PlayStation 5
Platforms


Personal info
Jisoo Kim
From South Corea - Busan
Age: 37
Occupation
Beautician
Technology and Videogames Knowledge
Technology: 2/5
Videogame: 3/5
Goal
Jisoo likes to manage important assets, and collect all that's possible
Background Description
Jisoo Kim works as a beautician in a small salon for several years. Outside of work she finds comfort in calm evenings at home, where she plays 2D games enjoying social interactions.
Bartle's Type
Socializer
Achiever
Criteria for Videogames
She loves 2D graphics, and cute animals/beings
Platforms
Windows
Frustration with Videogames
She hates to play Videogames with anti-aesthetic visuals

4. Legal Analysis

‘Hivemind’ is an IP created by **‘Tri-Side Die’** software house. On TMView, the software house’s name appears to not be registered.
All of the assets for the game are entirely made by the team or commissioned to external artists as specified in the TDD document.

The game features some scenes that may be frightening to children, considering it contains depictions of insects, even if pixelated. The game also contains non-realistic depiction of violence and may contain online interaction as it is a multiplayer game. The game will contain the option to purchase digital goods such as expansions later down the production line.
Taking into consideration all of this, we expect to receive a PEGI 7+ rating for our game.





Icon for the game, and socials



Company logo



Game title



Game title - pixel art

5. Gameplay

5.1 Overview

All specific values, when it comes to all gameplay resources like time, drop rates and so on, are subject to change after playtesting. The game aims at a challenging experience that keep the player entertained without frustrating them with

5.2 Gameplay description

5.2.1 Hive storing system

Before explaining the various mechanics of the game it is important to understand how the data relative to the hives are saved.

All the information about a player hive is saved inside a HiveRegistry. The HiveRegistry maps a ulong which is the unique id of the player to its HiveEntry inside the current session.

```
public static class HiveRegistry
{
   private static readonly Dictionary<ulong, HiveEntry> Registry = new();
    private static readonly Dictionary<Tilemap, ulong> TilemapOwnership =
new();
}
```

The **HiveEntry** class keeps everything important about a player's hive.

PlayerNumTag: is an enumerator that needs to be synced with other players to identify what is determining color.

HiveData is the list of rooms inside of a player's hive. Each room has a list of Vector2Int which is the list of positions occupied by the room itself. These cells are the values returned by the Unity grid that follow the OddQ coordinate system. This is important to mention since it is going to create a couple of problems with the hive merging system.

AxialOffset is a Vector2Int that represents what translation needs to be applied to hive when rendering it. This information is important for the hive merging system.

PlayerHive is just a collection of references to a player's Tilemaps.

```
public class HiveEntry
{
   public PlayerNumTag PlayerNumTag;
   public HiveData HiveData;
   public Vector2Int AxialOffset;
   public PlayerHive PlayerHive;
}
```

```
public class HiveData
{
   public List<HiveRoomData> rooms;
}
```

```
public class HiveRoomData
{
   public string id;
   public List<Vector2Int> cellPositions;
   public List<FurnitureData> furniture;
   public bool isPermaLocked;
   public bool isLocked;
```

```
}
```

5.2.2 Save and load system

When first entering the game, the player will not have any hive to play in, so it will need to create a new one. To create a hive the player simply needs to enter the hive name and the game will automatically generate the starting rooms. The system automatically generates a json file containing:

```
public class GameData
{
   public string id;
   public string name;
   public HiveData hiveData;
   public List<AvailableStructuresData> availableFurnitureData;
   public List<AvailableStructuresData> availableWorkbenchesData;
}
```

```
public class AvailableStructuresData
{
   public string id;
   public int count;
}
```

Id is a unique string generated to identify the hive created.
Name is the name chosen by the player for the hive created.
HiveData is the same structure mentioned in the previous section about hive storing system. Upon creation it is filled with the starting rooms specified in the starting template.
AvailableFurnitureData and **availableWorkbenchesData** are lists that contain respectively the unlocked furniture and workbenches and their corresponding available count. When creating a new hive, these lists are empty, since the player needs to obtain these structures.

When creating a hive, if not already present, it is created a manifest.json file that contains a list of hive summaries, which is a compressed list of all the hives created by the player.

```
public class HiveSummary
{
   public string id;
   public string name;
   public string createdAt;
}
```

Id is the unique string generated upon hive creation, it is the same id contained in GameData and it is used to link the two objects.
Name is the same as GameData.
CreatedAt is the date and time when the hive was created.

All this information is saved to be displayed in the hive selection screen. Once a player selects a hive, the system uses the id in HiveSummary to find the correct game data json with all the information required to load and display the correct hive.

5.2.3 Hive merging system

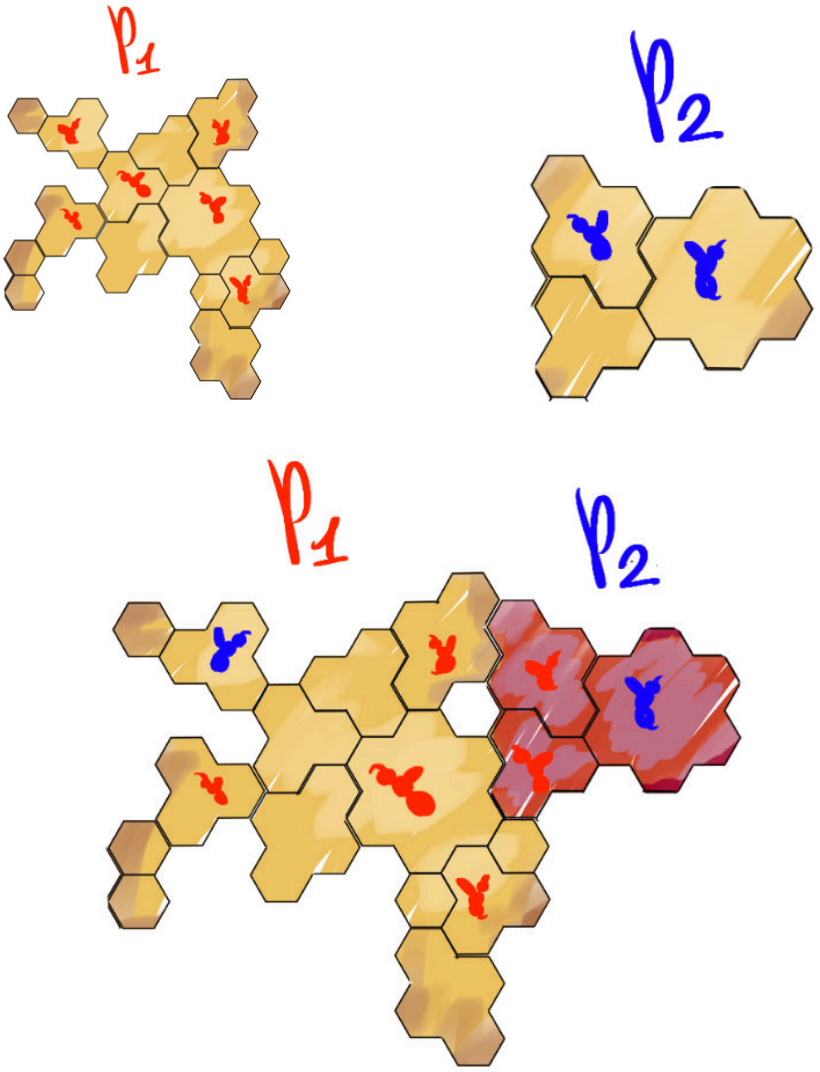
Now that we have a clearer picture of how our hive data is managed we can talk about the first important mechanic in our game, the hive merging system. This system is automatically invoked when a player joins another player's session and it is supposed to work independently of the number of players inside the session.

1. **Joining**: When a player joins another player's session he sends the host a json containing all of his hive's data.

2. **Data retrieval**: When the host receives the json it converts it back to HiveData and retrieves also the list of HiveData of the other players already in the session (his own HiveData included)

3. **Flattening and offset**: All the player's hives (except the new client) are flattened inside a single data structure, then the host computes the best position to place the new client's hive. The best position is computed by finding the position that maximizes the number of adjacent cells. This computation returns an axial offset. It is important that the offset is computed in axial coordinate space since a hexagonal grid in OddQ space cannot be consistently translated using a vector.

4. **Broadcast**: Once the computation is completed the result is broadcasted to the other clients. The clients that were already in the session receive the hive data to display and the correct offset, while the new client receives the data of every player in the session and their offset (himself included). This allows every player to have all hives data and positions synced.



Two players joining their hives together.

5.2.4 Editing system

This is one of the most important systems in our game since it allows players to expand and upgrade their hive. The system is divided into 3 different sub modes:

1. **Cell edit mode**:
   This mode allows players to edit the layout of their hive by painting new hexagonal cells or removing existing ones. All modifications are staged as pending changes, which means that the player can decide whether to either commit or revert them. In multiplayer mode the system follows the server authoritative paradigm, meaning that every modification request is to be

validated by the host before being returned to any client needing it.

2. **Furniture edit mode**:
   This mode allows players to choose between their available furniture and place them inside a room, providing buff to buffs to bees and/or workbenches. Just like cell edit mode it follows the server authoritative paradigm.

3. **Workbenches edit mode**:
   This mode works the same way as the furniture editing system, but instead of placing furniture the player places workbenches that can be used by bees for their chores.



In order from top to bottom: Cell Edit, Furniture Edit, Workbenches Edit.

5.2.5 Cell editing system

In this section we look at a more in depth description of how the cell editing system works.

5.2.5.1 Cell states

First of all we need to specify that in cell edit mode, the cells of the hive can be in different states, depending if they have been touched by the modifications or not.

- **Confirmed cells**: A cell that has been committed to the hive, it is owned by a specific player and persists across multiple sessions unless deleted.

- **Pending addition**: A cell that a player has marked as an addition during the editing period but has not been confirmed yet. It is visible to other players in edit mode as a translucent overlay with the color of the player that cell is being built for.

- **Pending deletion**: A confirmed cell that a player has marked for deletion but has not been committed yet. This cell is visualized only to players in edit mode with a slightly red translucent overlay.

- **Locked cell:** A confirmed cell that cannot be deleted unless turned back into a confirmed cell.

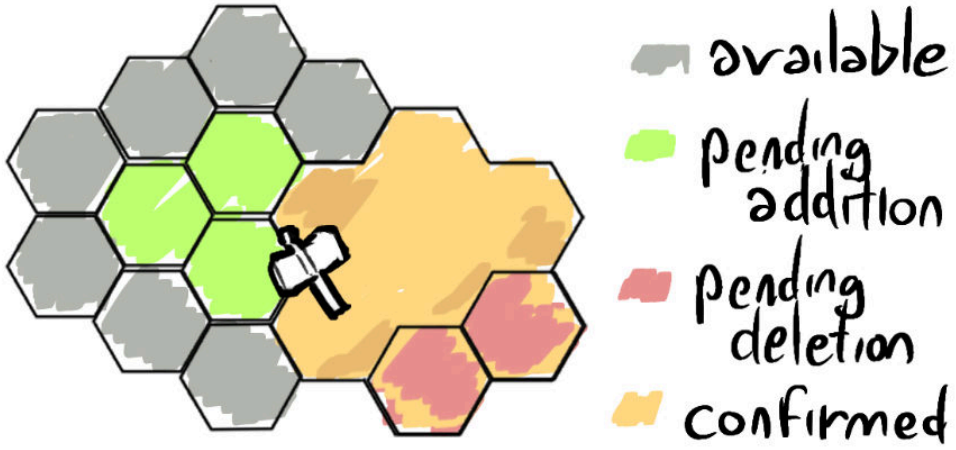
- **Perma-locked cell**: A locked cell that can never be turned into confirmed cells. Usually they are generated when creating the hive the first time and they include the cells forming the starting rooms. Perma-locked cells cannot be generated voluntarily by the player.

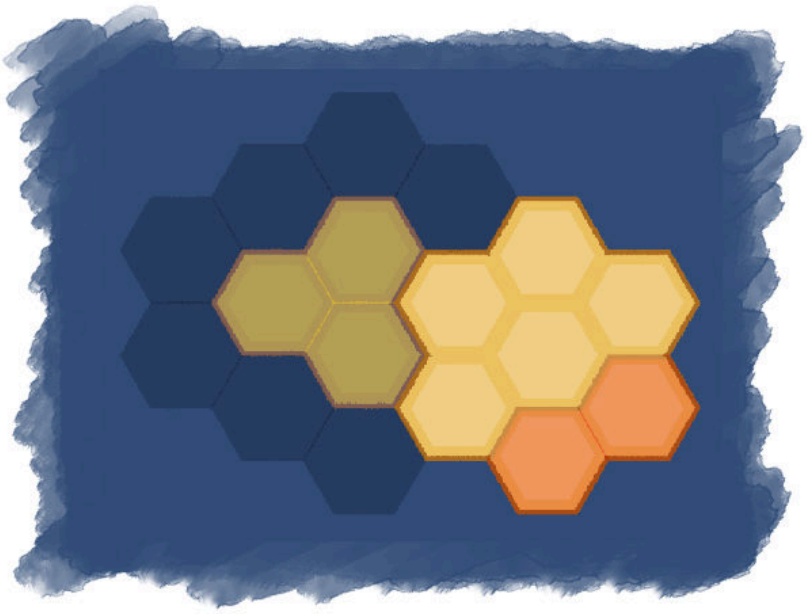
5.2.5.2 Spatial sets

To better understand how the system works, it is important to distinguish the various sets used to divide the cells during edit mode.

- **Available cells**: The set of cells that are valid starting points for building new cells for a targeted player.

- **Paint frontier**:The set of empty cells adjacent to at least one pending addition being staged by a specific builder. It represents the cells that can be painted next in the current building stroke.





5.2.5.3 Player roles

In cell edit mode there are two main roles:

- **Builder**: The builder is the player actively in cell edit mode, making changes to the merged hive.

- **Building target**: Is the target chosen by a builder. A builder can choose a building target before starting to effectively build cells. Cells being built by a player are added to the HiveData of the building target.

5.2.5.4 System flow

Entering cell edit mode:
A player can enter edit mode by pressing the designated button in the UI or through a shortcut by pressing the “E” key on the keyboard. When entering edit mode, the player is shown the current progress of other players currently in edit mode.

Switching building target:
When entering edit mode, cell edit mode is the sub mode automatically selected. The player is shown the available cells for the currently selected building target, that is himself when entering the mode. With scrollwheel the player can switch the building target among the players inside the session.
Players can switch building targets only when the builder has no pending modifications, meaning that both the list of pending additions and deletions need to be empty.
Switching between different building targets shows the available cells of the target. This set needs to keep in consideration also the pending additions and deletions of other players that might influence the currently selected target.

Adding pending cells:
A player can add pending cells by pressing the left mouse button while hovering a cell inside the available cells of the building target. After the first cell has been placed the paint frontier is computed, and the builder can only place additional cells while pressing LMB and hovering on a cell belonging to the paint frontier. Each newly painted cell updates the painting frontier.
Adding a pending cell influences the builder’s own paint frontier but it might also influence the available cells and painting frontier of other players.

Removing confirmed cells:
A player can mark cells as pending deletions by pressing RMB while hovering a confirmed cell. The cell is sent to the host to confirm that the cell can be actually marked as a pending removal (the checks to be performed for the deletion of a cell are going to be explained in a separate section). A builder can mark cells as pending deletions independent of the current building target.
Marking confirmed cells as pending deletions might influence other players' available cells.

Unmarking pending deletions:
A player can unmark cells from pending deletions by pressing LMB while hovering on a cell marked as a pending deletion.
Unmarking cells marked as pending deletions might influence other players' available cells.

Unmarking pending additions:
A player can unmark cells from pending additions by pressing RMB while hovering on a cell marked as a pending addition. A player can unmark only its own pending addition and not the ones of other players.
Unmarking cells marked as pending additions influence the builder’s own paint frontier and might also affect other players' available cells and painting frontiers.

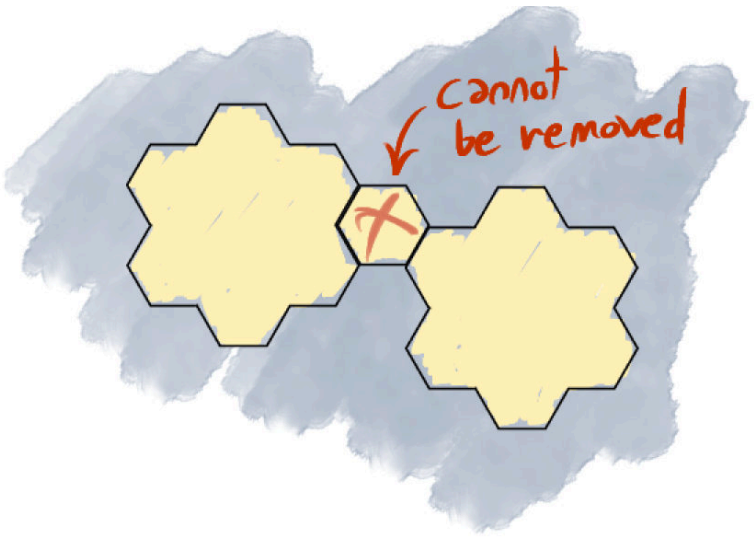
Committing changes:
When a player is satisfied with the changes made, he can decide to commit the changes. Cells marked as pending additions become confirmed cells and they are saved inside a single room (this happens because of the connectivity checks imposed by the system). Cells marked as pending deletions become empty cells. After this action is completed the player remains in cell building mode, and can go on with building new rooms.

Reverting changes:
The player could also decide to discard the changes made and empty the list of pending additions and deletions. After this action is completed the player stays in cell building mode and can start making changes again, and since the pending modifications lists are empty he can also change the building target.

5.2.5.5 Connectivity checks

When a player tries to delete a cell the system must ensure these things:

- **Merged hive connectivity**: The graph of the merged hive, containing the hives of all players in the session, has to remain connected, meaning that every cell must have a path to reach every other cell in the merged hive.
- **Individual hive connectivity**: Since a player might decide to play with his hive in mode single player and multiplayer mode, his hive needs to remain connected in both instances. This means that the same connectivity reasoning that was applied for the merged hives need to be also applied to every single hive individually.
- **Room connectivity**: The same connectivity rules of the hives must be also applied to individual rooms, which means that a deletion cannot split a room into two different rooms.



To speed up the process of checking if a deletion maintains connectivity in these three different domains we simply consider the 6 neighbouring cells and apply Breadth First Search to ensure that these six cells are connected among themselves. This approach works since before each deletion the hive is connected. This way we avoid checking every single cell and speed up the computation time significantly.

5.2.6 Multiplayer vs Singleplayer

The editing system doesn't have any particular change when switching between singleplayer and multiplayer. The only change is the fact that in multiplayer, players cannot see the changes being made by other players in edit mode unless they are also in edit mode. Only once the changes have been committed also the players not in edit mode receive the hive changes.

To enhance the feeling of playing together and make communication between players easier, all cursors will be visible to all players in the game world. At any time every player can tell what someone else is looking at, what they're working on, and if they are in building mode or not. To do this we synchronize all the player cursors positions using the global positions in the world rather than on their screen.



Slay The Spire 2 - All player cursors are shown during gameplay

5.2.7 Resources

Hivemind, as an artificial life simulation game, wants to imitate the real world, which is characterized by resources (among other aspects). For this reason our bees need the following elements in order to sustain the colony: pollen, water, resin, nectar, honey, food and glue. Every resource's utility is based on real world dynamics:

- Pollen *"is a vital resource for bees, providing essential proteins and nutrients needed for growth, development, and overall colony health. In addition to being consumed by adults and larvae, it also plays a key role in hive construction by contributing to the materials used in building and reinforcing the structure. It supports both brood development and the stability of the colony's living environment."* Like aforementioned, pollen is a vital factor for the colony's structure, so we decided to ignore the aspect about nutrient properties, and to implement this element in Hivemind's resources system.

- Water *"Water is essential for bees to regulate hive temperature and maintain hydration within the colony. They collect and distribute it inside the hive, using it to cool the nest and dilute food for larvae and adults."* For this reason we thought to adopt the water resource as a crafting component for food (crafted resource).

- Resin *"Resin is a key material collected by bees from plants and trees, used to produce propolis. They use it to seal gaps, reinforce the hive structure, and protect the colony from bacteria and external threats."* According to this, resin is an optimal resource for repairing/reinforcing the hive structure, so mixing resin and pollen returns glue (crafted resource).

- Nectar *"Nectar is a sugary liquid collected from flowers and is the primary raw ingredient for honey production. Bees transport it back to the hive, where it is processed and stored as a long-term energy source for the colony."* As everyone knows, nectar is the main ingredient for honey production.

- Food is a crafted resource, and is very important for bee colony survival. This resource is used by the bees automatically, without player involvement. When a bee wants to eat, the game checks if the amount can be subtracted from the resource storage, if possible the amount consumed appears over the bee and it starts an eating animation.

- Glue is another crafted resource, obtained by mixing resin and pollen. This substance is used to expand the hive (by building new hexagonal units). Once building mode changes are confirmed, the total cost of the applied modifications is deducted.

- Honey is the quintessential element when it comes to bees. It is the final crafted resource, and it is obtainable by refining the nectar into the furnace. This resource is used as primary currency in the game.

This table represents some resource's dynamics.
All the values are subject to change, according to play testing

Food consumption values for every bee

Bee Type	Consumption	Bee Type	Consumption
Honey bee	10	Killer bee	40
Bumble bee	15	Japanese bee	25
Carpenter bee	30	Sweet bee	35
Cuckoo bee	20	Leafcutter bee	40
Digger bee	20	Vulture bee	25

Furniture costs examples

Furniture	Honey	Water	Glue	Pollen	Nectar	Resin
Wax Lamp	10	-	10	-	-	10
Floral Wall Pattern	-	5	-	30	27	-
Honey Drip Decoration	24	-	-	-	12	-
Resin Sculpture	-	-	33	-	-	47
Pollen Carpet	14	-	-	25	-	-
Royal Hive Ornament	35	20	13	22	-	11
Crystal Nectar Droplet	-	28	-	-	43	-
Golden Honey Crest	45	-	-	18	-	10
Resin Glow Vines	-	10	26	-	-	30
Blooming Hive Mural	53	-	-	-	17	-
Queen's Emblem	100	-	25	9	-	36

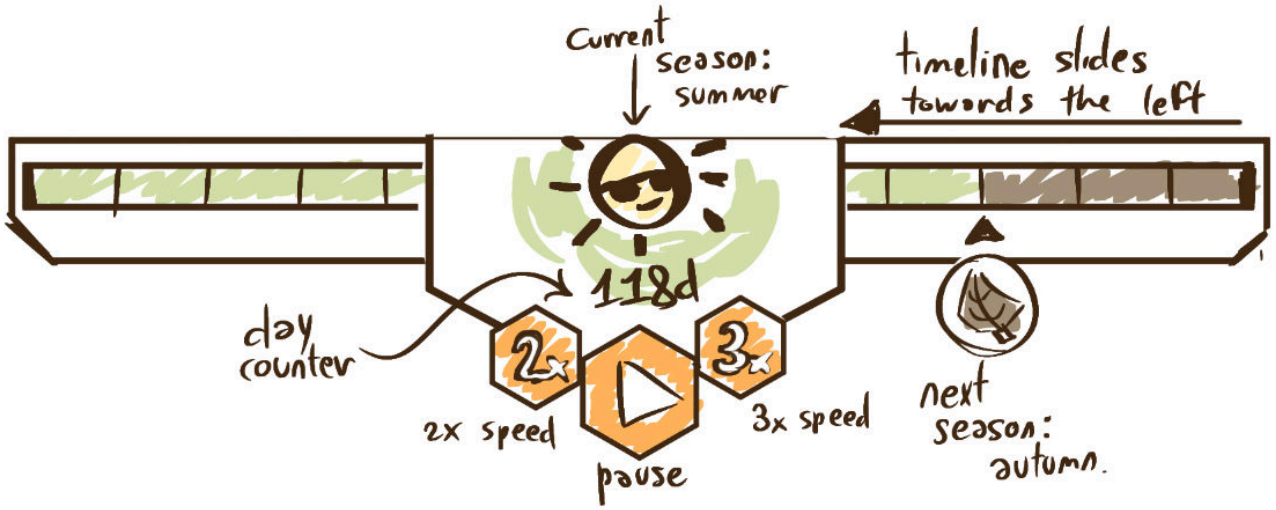
5.2.8 Time

Time is also another important 'resource' of the game. Completing tasks will take different amounts of time depending on the bees, as will sending a bee out foraging, waiting for larvae to grow into bees, and even how many days a bee should live!

The game takes inspiration from the Sims, using a time system that keeps a 1 game minute: 1 real-life second ratio, meaning a **full day**, in hive mind will take up 24 minutes. This information is important when it comes to the rest of the time sensitive mechanics the game has.

Additionally, as specified in section 7, the game will present seasons with their own gameplay challenges. Because we wish the player to experience all the challenges the game offers in a timely manner, we decided to have each season to last 60 days, meaning each season will last 24 real-life hours, and an entire year cycle in HIvemind will last 96 hours.

Of course, players will also be free to speed up the time rate to 2x or 3x as fast, meaning at max speed a player can go through an entire year in the game in 32 hours of gameplay. This feature allows players to relieve some of the tediousness of time consuming tasks, and gets them moving faster towards experiences they are more excited about such as events.



Time is represented in the game via a **timeline** at the top of screen, showcasing events that are incoming, how many days have passed in the hive, and the changing of seasons. The time line shows the 5 upcoming days, and the previous 5 days, the current day counter and season, and the aforementioned buttons to speed up and pause the game. While the game is unpaused the timeline slides towards the left showing the constant passage of time, and giving the player an idea of how rapidly the future is approaching even at faster speeds.
When players are playing in a multiplayer lobby, everyone has the same timeline as the host. When players return to their singleplayer world, their timeline will be restored to where it was before they left but they'll carry the progress they made in multiplayer with them.
This makes playing multiplayer an effective strategy in delaying events players are not ready to face, intentionally so as to invite the player to seek out friends to play with should they be in trouble.

5.2.9 Events

Events in our game are specialized scenarios that can occur randomly during the course of the game that challenge the player with unique situations in different ways as to test the strength and variety of their hive.
The game presents two types of events, each having different mechanics attached to them:

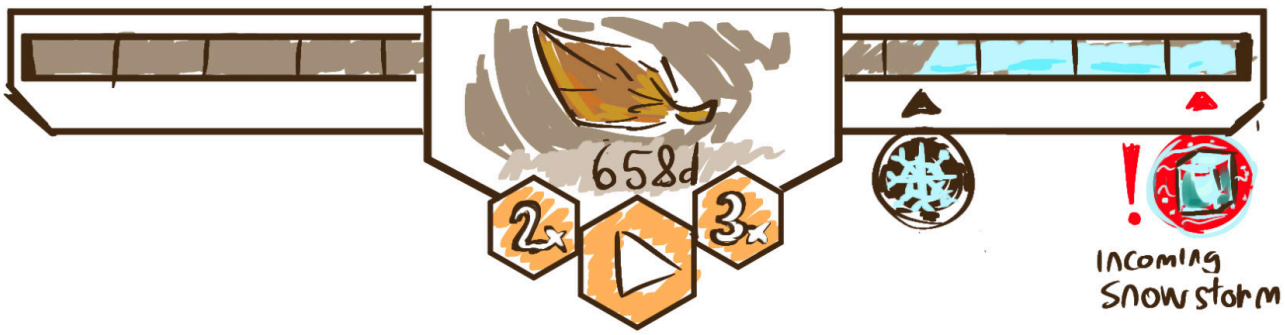
Hive Events
These occur at any moment during gameplay and directly affect the home of the player. Players will be able to respond to this using the bees they have inside of their hive. These events are aimed at slowing

down the progress the player is making and invite the player to specifically build their hives in ways that can oppose them or suffer greater consequences.
Every day, the game randomizes whether an event should spawn or not. If an event is spawned, it will be placed in the game's timeline to warn the player before it occurs, with the exception of the **Wax Moth** event as specified in section 6. An event is always spawned **5 days** after the current day. The game checks what season it will be when the event spawns, and picks a possible event from that season, depending also if the rules for the event are met, such as the Giant Hornet Attack.

Spring	Summer	Autumn	Winter
Pollen frenzy	Giant Hornet Attack	Flood	Snow Storm
Smoke bomb	Heat Wave	Wax Moth Invasion	Thermal Leakage

For more information regarding the events, read sections 6.2 and 7.3

Once an event has spawned, **no more events can be spawned until the coming one has been resolved**, this is to avoid overwhelming players with too many threats at the same time and give them more control over their hive to be able to restructure and prepare for the incoming threat.
Once the event has resolved, there'll be a period of 5 more days during which the game does not check if an event should be spawned, to give further room to breathe to the player.



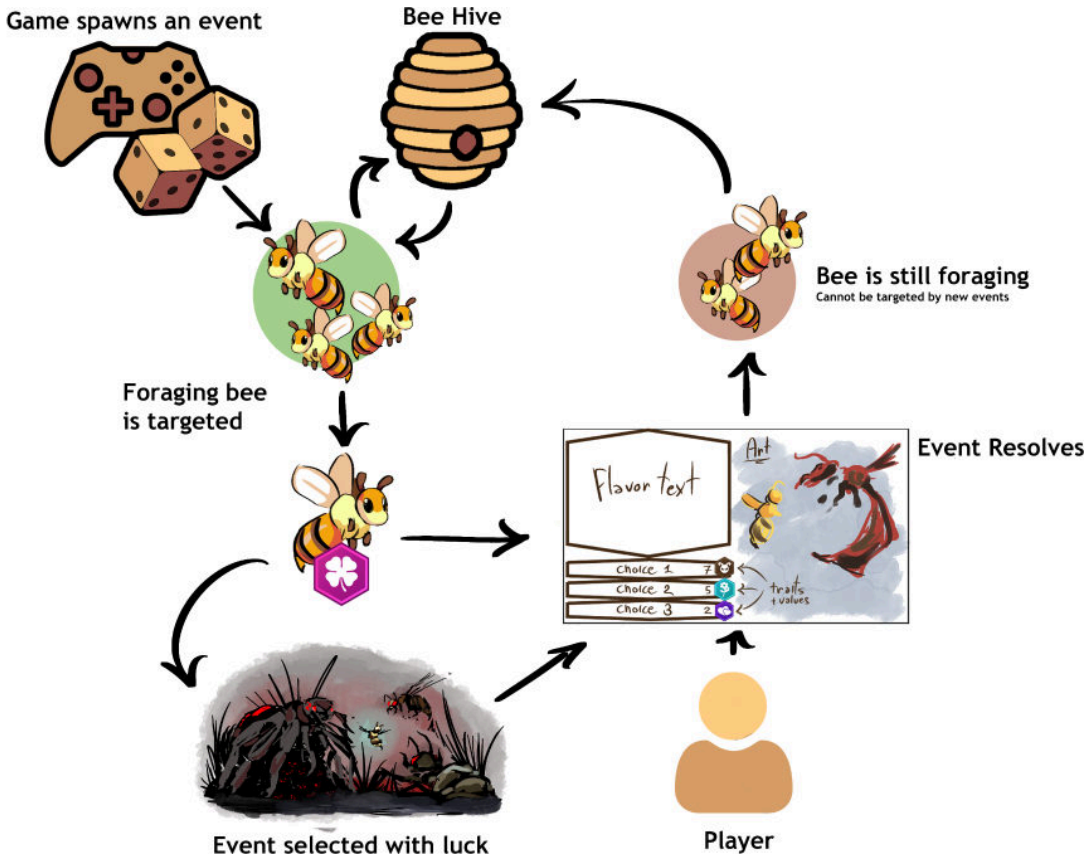
Once the 5 days are over the game returns to randomly spawning an event. Meaning in the worst case scenario where the game decides to spawn an event at every possible opportunity, the game can spawn 6 events every season.
Obviously this number is unrealistic, as the chance for an event spawning is reasonably low enough to avoid tediousness and too much repetition, considering the game only has 8 hive events planned on release.

In multiplayer, because everyone has the same timeline as the host, everyone will have to work together to defend against the events. This is why these events tend to scale with the number of players connected to a lobby, to avoid making these challenges too trivial with enough players.

Foraging Events
These occur to singular bees while they are out foraging. They propose interesting situations that a single bee will have to respond to using exclusively their own traits, as they are detailed in section 6.2.

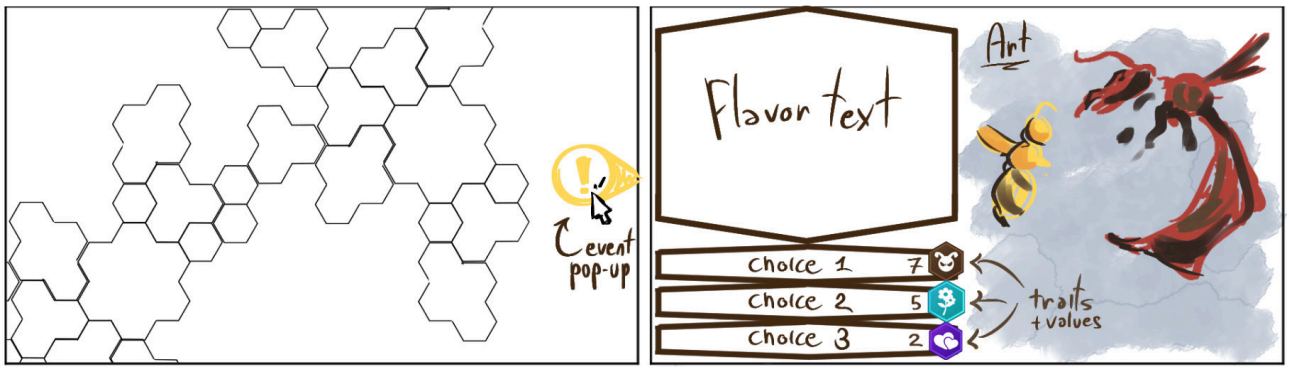
While any bee is out foraging, the game will attempt to spawn an event every 1 minute, picking a bee at random as its target for the event if it is spawned successfully. Once a bee has endured an event, they can no longer be targeted for the remainder of their foraging session, with the exceptions of **Japanese Bees** as detailed in section 6.2.
If no bees are out foraging, or there are no more bees sent out that have not endured an event, the game stops trying to spawn them.
The selection of the event depends on the bee's '**Luck**' trait and the event's difficulty. A higher luck trait means the pool of events the bee can be targeted by will be easier to overcome.

If a new bee is sent out in these conditions, the game will wait 1 minute before it tries to spawn the first new event.



When a foraging event occurs, the player is notified in the notification section of their UI. The player can then press on that notification to be presented with the event in question. The event interface is composed of:

- Flavor Text: Detailing the context of the event, the bees current situation, and setting the mood for the scenario
- **Choices:** 3 clickable options containing flavor text explaining what the bee should attempt to do to resolve the event, and an icon showcasing the trait the bee will attempt to use for that outcome
- **Art:** A simple illustration that showcases the flavor text to draw in the player

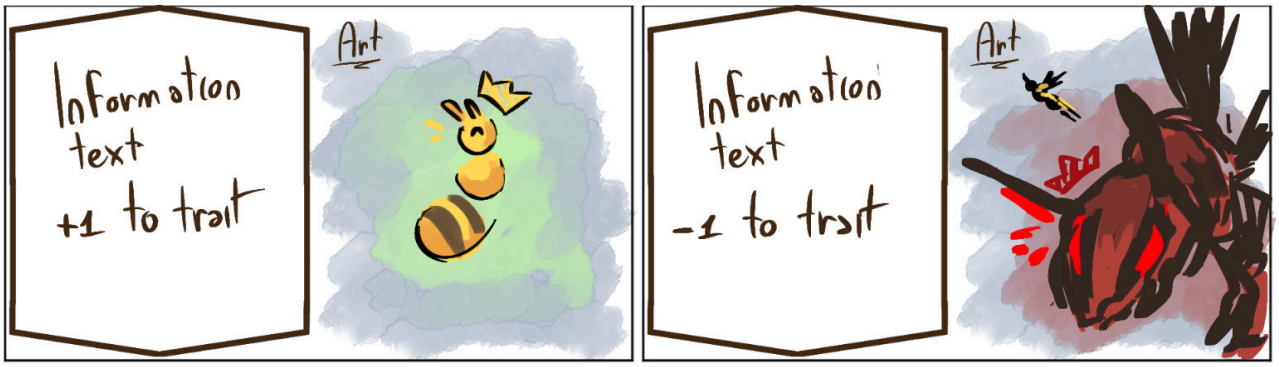


Inspired by tabletop games, each event has a hidden **Target Difficulty**, meaning how difficult it will be to succeed in the event; This is also used in combination with the Luck trait to decide which event should target the bee in the first place. When the player picks a choice the game first randomly generates a number within a specific range and then adds the trait value of the bee onto the number 'rolled' by the

game. If the total value is equal or greater than the Target Difficulty, then the positive outcome is given to the player, otherwise a negative outcome will occur instead. The system is simplified and uses mechanics a player is generally expected to be familiar with, given our target audience, to avoid once again potentially overwhelming the player with too many decisions.

Once an event resolves, the outcome interface will show:

- **Information Text**: In part flavorful detailing the result of the event story wise, but also explaining the positive or negative outcomes that will reflect on the gameplay.
- **Art:** A simple illustration that showcases the result of the event.



In multiplayer, each player will be responsible for their own bees, meaning each player can only respond to their own events, and their events will not show up to other players in the lobby.

Finally, certain foraging events cause player bees to gain **mutations**, or to find new friendly **bees of different species** to bring to their hive, all of which are showcased in section 6.2, opening the doors to gene mixing and the breeding mechanics of the game, to create stronger and more resourceful bees in the hive.

In summary, these two event types target players in different ways and propose to them challenges that need to be tackled with very different approaches. The first rewards a powerful hive, strategic building, and social play, while the second rewards experimentation, and creation of singularly powerful bees.

5.2.10 Breeding

To start creating more bees, players will need **nurseries**, workbenches where the queen can deposit larvas periodically, and where bees can add their own genes to the larva for a chance to create something new and exciting.

Every 6 minutes, the queen bee will first check if there is any available space inside of any nursery, which can contain a maximum of 4 larvae. Then the queen bee checks if should lay an egg by taking into consideration; Amount of food in the hive, consumption rate of bees, number of bees in the hive, and production rate of food. If the first condition is met, and the second check returns positive, the queen bee will lay an egg in a nursery and repeat the process after 6 minutes.

The computation done for the second condition won't be perfect, and isn't expected to be so. It's a way to make the player's avatar more resourceful on its own but not artificially perfect, even queen bees are allowed to make mistakes!

Once an egg has been placed in a nursery, other bees can interact with it to fertilize it with their species makeup! Which will determine what the fully grown bee will look like at the end of the growth period.

Bees can do this on their own periodically, according to their 'Sweetness' trait, which adds a chance for the bee to try and add their genes to a larva on their own, but can obviously also be player controlled. The reason for this is to avoid excessive player min-maxing of exceptional bees and also to enhance the artificial life feeling of the bees. The player only has partial control over their actions.

An egg takes 3 days to mature into a full bee. Whenever a bee has fertilized an egg, they cannot do so for another day, to avoid having single bees insert their genes in every possible egg.

The actual breeding process involves a lot of percentages, and is best showcased using a series of examples of increasing complexity. But before that some important information:
The **species make-up** of a bee is determined by its three parts (head, body, stinger). A bee that has all parts of the same species is considered as 100.00% of that species (pure honey, bumble, carpenter bees…). With bees that are mixed the percentages are broken up as one would expect: A bee that has the **head of a bumble bee**, and the **body and stinger of a carpenter bee** for example, will be **33.33% bumble bee**, and **66.67% carpenter bee.**

When an egg is laid, it is considered initially to be 100% a honeybee, this is because the queen of the hive is always a honeybee, but also to make sure some honeybee genetics are always added into the mix, once again to avoid players excessively min-maxing their best bees with very little effort, it is still possible but requires more involvement.

Now, here is the breakdown of what happens when bees start fertilizing the egg:

Example 1: Egg + 1 Pure Bumble Bee

Bee	Species make-up	Percentages breakdown
Egg		100% Honey Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Resulting Larva	1/2 * (100.00) + 1/2 * (100.00) = 50.00 + 50.00	

This first example is quite simple. When a pure bumble bee fertilizes an unfertilized larva, the larva will be half honey bee and half bumble bee! The fractions represent the weight that each contribution has on the final larva, since everything has the same weight it will trivially be $1/numOfContributingBees$, which was color coded to ease readability. The numbers between parentheses represent the color coded **Percentage breakdown** of the bee, which currently is simply 100.00 since we are using pure bees.

Example 2: Egg + 1 Pure Bumble Bee + 1 Pure Carpenter Bee

Bee	Species make-up	Percentages break down
Egg		100% Honey Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Carpenter Bee	Carpenter Bee Head Carpenter Bee Body Carpenter Bee Stinger	100% Carpenter Bee
Resulting Larva	1/3 * (100.00) + 1/3 * (100.00) + 1/3 * (100.00) = 33.33 + 33.33 + 33.33	

This second example continues with the simplicity of pure bees. If a Carpenter bee adds their genes to the egg now, then the resulting larva will simply be split among the three contributions, an even 33.33% split.

Example 3: Egg + 2 Pure Bumble Bees + 1 Pure Carpenter Bee

Bee	Species make-up	Percentages break down

Egg		100% Honey Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Carpenter Bee	Carpenter Bee Head Carpenter Bee Body Carpenter Bee Stinger	100% Carpenter Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Resulting Larva	1/4 * (100.00) + 1/4 * (100.00) + 1/4 * (100.00) + 1/4 * (100.00) = 25.00 + 25.00 + 25.00 + 25.00 = 25.00 + 50.00 + 25.00	

Now, if a second bumble fertilizes the egg, the genes will stack in the total, meaning bumble bees make up effectively half of the genetic make-up of the larva, while the remaining fourths remain of honey and carpenter bees.
Things grow more complicated once we aren't dealing anymore with just pure bees, meaning the percentages breakdown will contain different species! As shown in the next example.

Example 4: Egg + 2 Pure Bumble Bees + 1 Pure Carpenter Bee + 1 Mixed Bee

Bee	Species make-up	Percentages break down
Egg		100% Honey Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Carpenter Bee	Carpenter Bee Head Carpenter Bee Body Carpenter Bee Stinger	100% Carpenter Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Mixed Bee	Carpenter Bee Head Carpenter Bee Body Digger Bee Stinger	66.67% Carpenter Bee 33.33% Digger Bee
Resulting Larva	1/5 * (100.00) + 1/5 * (100.00) + 1/5 * (100.00) + 1/5 * (100.00) + 1/5 * (66.67 + 33.33) = 20.00 + 20.00 + 20.00 + 20.00 + 13.33 + 6.67 = 20.00 + 40.00 + 33.33 + 6.67	

In this case as we can see, the mixed bee's genes are diluted by all the other species present in the larva. Since its contribution will only count for 1/5th of the whole larva, its 33.33% digger bee becomes 6.67% in the larva.

Example 5: Egg + 2 Pure Bumble Bees + 1 Pure Carpenter Bee + 1 Mixed Bee + 1 Mutated Mixed Bee

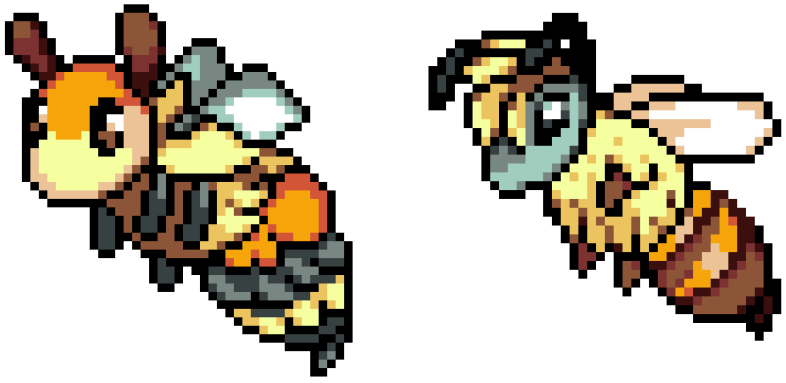
Bee	Species make-up	Percentages break down
Egg		100% Honey Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Carpenter Bee	Carpenter Bee Head Carpenter Bee Body Carpenter Bee Stinger	100% Carpenter Bee
Bumble Bee	Bumble Bee Head Bumble Bee Body Bumble Bee Stinger	100% Bumble Bee
Mixed Bee	Carpenter Bee Head Carpenter Bee Body Digger Bee Stinger	66.67% Carpenter Bee 33.33% Digger Bee
Mixed Bee	Angler Head Digger Bee Body Cuckoo Bee Stinger	33.33% Angler Mutation 33.33% Digger Bee 33.33% Cuckoo Bee
Resulting Larva	1/6 * (100.00) + 1/6 * (100.00) + 1/6 * (100.00) + 1/6 * (100.00) + 1/6 * (66.67 + 33.33) + 1/6 * (33.33 + 33.33 + 33.33) = 16.67 + 16.67 + 16.67 + 16.67 + 11.11 + 5.56 + 5.56 + 5.56 + 5.56 = 16.67 + 33.34 + 27.77 + 11.12 + 5.56 + 5.56	

Lastly, as this example shows, mutations have the same split as species, and how when adding many different mixes things can get quite unpredictable. However, the final **species make-up** of the larva, can only be made up of a maximum of three bee parts, so how is that established?

Firstly, to avoid any cascading errors due to the two point decimal precision we are using, we normalize the percentages over 100. Then, **for every part of the bee**, the game determines what species that part should be, or if it should be a mutated part if present in the larva's genetics, according to the normalized percentages.

So, following our last example, when the larva grows into a full bee, the head will have a 16.67% of being a honey bee's head, 33.34% of a bumble bee, 27.77% of a carpenter bee, 11.12% of a digger bee, 5.56% of being a mutated angler head, and 5.56% of a cuckoo bee.

The same reasoning is then done independently for the body, then the stinger, and finally our new larva has grown into a never-seen-before bee!



A bumble bee with a honey bee head, and a honey bee with a bumble bee head!

More information about how the bees traits are decided, and what does traits are can be found on section 6.2

5.3 Controls

This section describes the flow of the player's interactions and key bindings.

5.3.1 Bindings

The control mapping tables have been created by considering the Unity Action Maps and the context in which they have to be enabled.

Game context:

Action	Keybind	Notes
Enter Edit Mode	E	Enables edit mode context and opens the UI with the 3 submodes. By default cell edit mode is enabled meaning that also its context is enabled.
Open Chat	T	Enables chat context and opens the chat's UI. Disables the current enabled contexts.
Interact	LMB	Interacts with various elements in the scene, such as bees, used to assign tasks to them. Also used to interact with workbenches. Also selects and puts in focus a bee when interacted with.
Pan Camera	RMB	Allows the player to move the camera around the game scene.
Zoom In	Mouse wheel up	Allows to zoom in the camera for a closer look into the hive
Zoom out	Mouse wheel down	Allows to zoom out the camera for a whole view of the hive.

Chat context:

Action	Keybind	Notes
Close chat	Esc	Disables chat context and closes the chat's UI. Enables the contexts that were enabled when the chat was opened.
Send message	Enter	Send a message to all the players in the session.
History up	Up arrow	Set the current message to be sent to the previous message in the history.
History down	Down arrow	Set the current message to be sent to the next message in the history.

Edit mode context:

Action	Keybind	Notes
Exit Edit Mode	E	Disables the edit mode context and the currently enabled specific edit mode context. Also closes the edit mode UI.

Cell edit mode context:

Action	Keybind	Notes
Paint	LMB	Tries to mark a cell as pending addition or unmark a cell marked as pending deletion.
Delete	RMB	Tries to mark a cell as pending deletion or unmark a cell marked as pending addition.
Confirm	Enter	Submit the changes to the merged hive. Deletes the cells in pending deletions and adds the cells in pending additions as a single room.
Cancel	Esc	Discard the changes saved in pending additions and deletions.
Scroll	Scroll wheel	Changes the building target of the player. Can scroll up or down to go to the next or previous player.

Furniture edit mode context:

Action	Keybind	Notes
Place	LMB	Tries to place the currently selected furniture inside a room in the position hovered by the cursor.
Rotate	R	Rotates the visual of the currently selected furniture.
Mirror	M	Mirrors the visual of the currently selected furniture.

Workbench edit mode context:

Action	Keybind	Notes
Place	LMB	Tries to place the currently selected workbench inside a room in the position hovered by the cursor.
Rotate	R	Rotates the visual of the currently selected workbench.

Mirror	M	Mirrors the visual of the currently selected workbench.

5.3.2 Procedures

Create new game:

1. **Authentication**: The player needs to authenticate via Unity Gaming Services to access Cloud Save and Relay features.

2. **Hive creation**: The player navigates through the UI and reaches the screen to create a new hive. Clicking on the button changes the scene to the one with the new hive.

3. **Allocation**: The player, that in this case is the host, sends a request for a relay allocation to the Unity Relay Service.

4. **Lobby Creation**: The host creates a Unity lobby and Steam lobby and stores the join code in the steam lobby metadata.

5. **Connection**: The host binds to the relay node and gets ready to receive RPCs from clients.

Join another player's session:

1. **Join game**: The player navigates through the UI and reaches the screen to enter one of his hives, select one and click play.

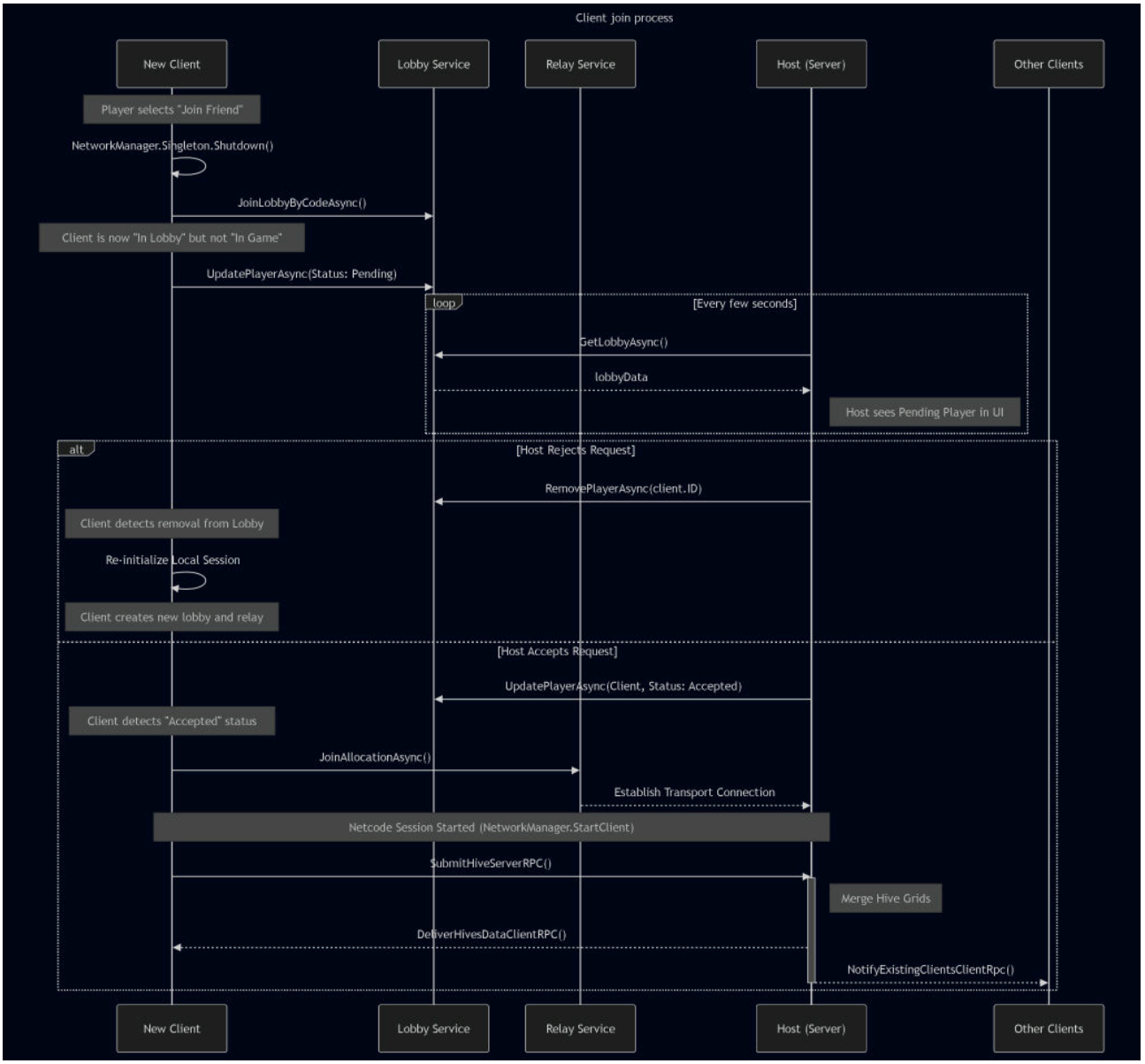
2. **Send request**: The player enters a friends room code and sends the request or uses the Steam overlay to check which one of his friends are playing the game and are in a session he can join, then sends a request. When sending a request the client needs to connect to the same session of the host, meaning he has to disconnect from his own NGO connection and temporarily join the one of the player he is sending the request to.

3. **Handle request**: The host, every 2 seconds, polls from the lobby data to check for incoming requests. When he finds a request he can decide whether to accept or reject it.
    a. **Reject**: The client sending the request is sent a notification about the rejection and then is kicked from the host session. At this point he needs to create a new session for himself.
    b. **Accept**: The client sending the request is notified about the accepted request.

4. **On request accepted**: The client sends over the network his hive data.

5. **Host computation**: The host receives the new client's hive data and computes the optimal position based on the position of the hives of all players in the session, and then broadcasts the computation to all clients.

6. **Broadcast**: The new client receives the hive data of all the players in the session and renders everything. The other clients receive the hive data of the new client and its position and then render it.



Add cell to pending additions:

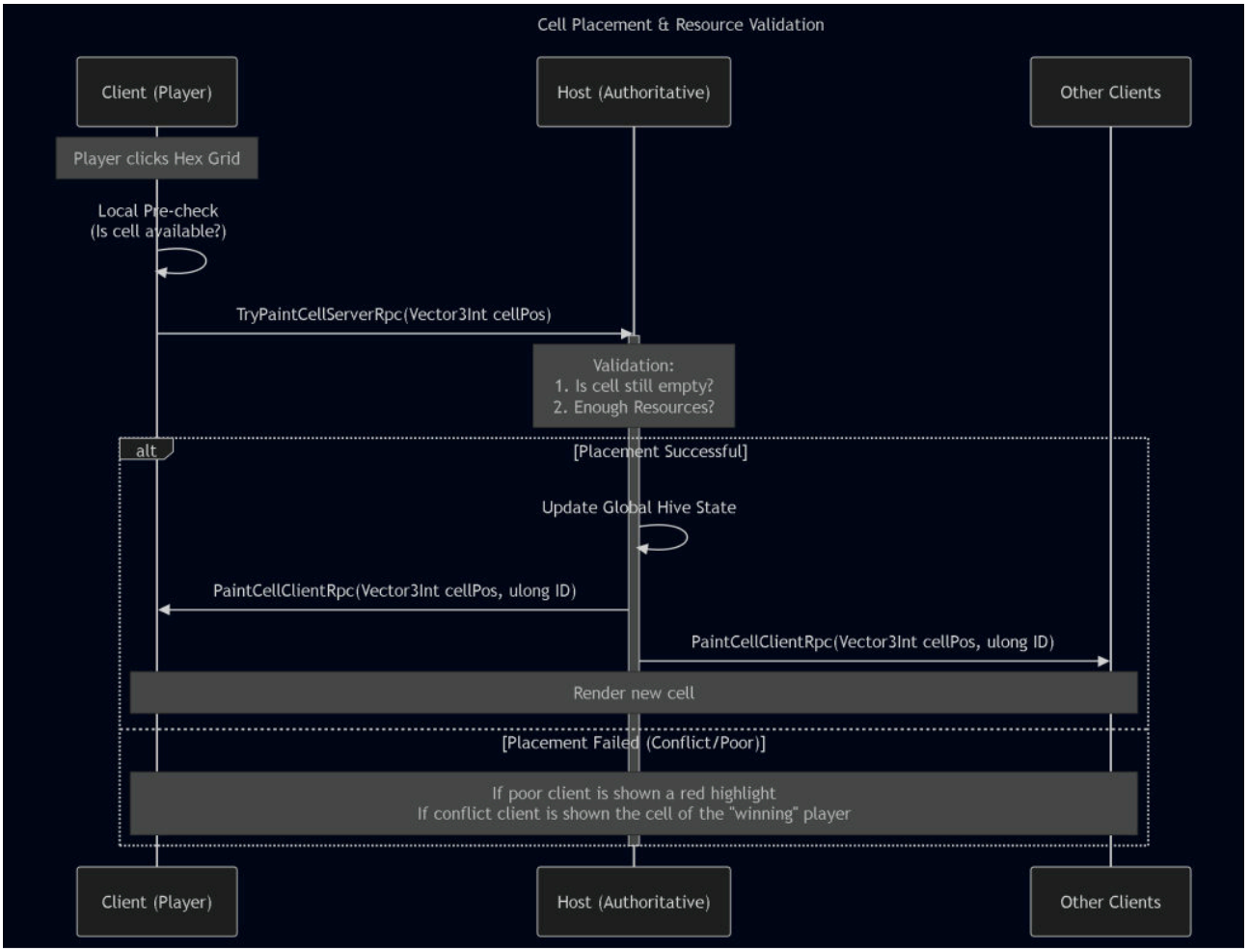
1.2.2 **Enter edit mode**: A client presses E or the specified button to enter edit mode and unless already selected, he selects the cell edit mode.

2.2.2 **Select building target**: The client uses the scroll wheel to switch to the desired building target.

3.2.2 **Click on a cell**: The client clicks on a cell with the left mouse button and sends a ServerRPC.

4.2.2 **Host evaluation**: The host receives the RPC invocation and checks if the cell can be marked as a pending addition.

5.2.2 **Feedback to client**: The host sends the result of the computation back to the builder and if the placement was a success it is also broadcasted to all other clients if they are in edit mode.



Note: The same procedure is applied for marking a cell as a pending deletion, unmarking a pending deletion and unmarking a pending addition, with the only difference that the host has to perform different checks during the computation and that what is rendered as a result depends on the initial action performed.

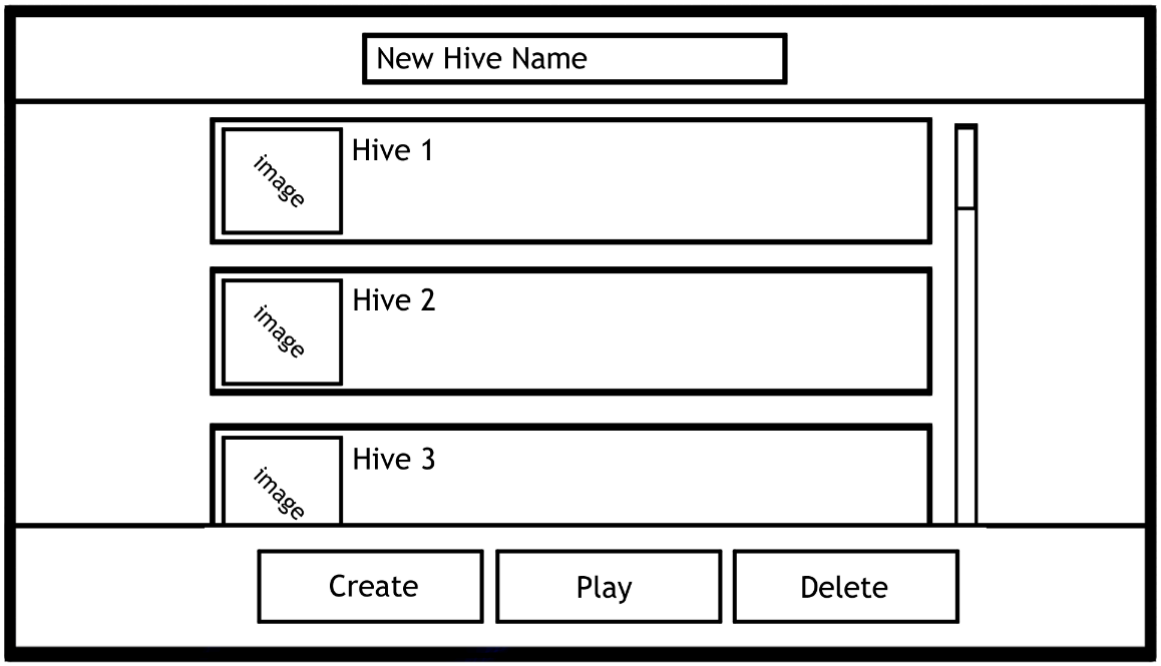
5.3.3 Interfaces

Main Menu



This is the first interface of the game,
Play leads to the hive selection screen, where the player decides which one of their hives to load from memory and then play, or to create a new hive from scratch, or delete an existing hive.
The **Settings** allow the player to change graphics and audio.
Quit exists the application.

Hive Selection Screen



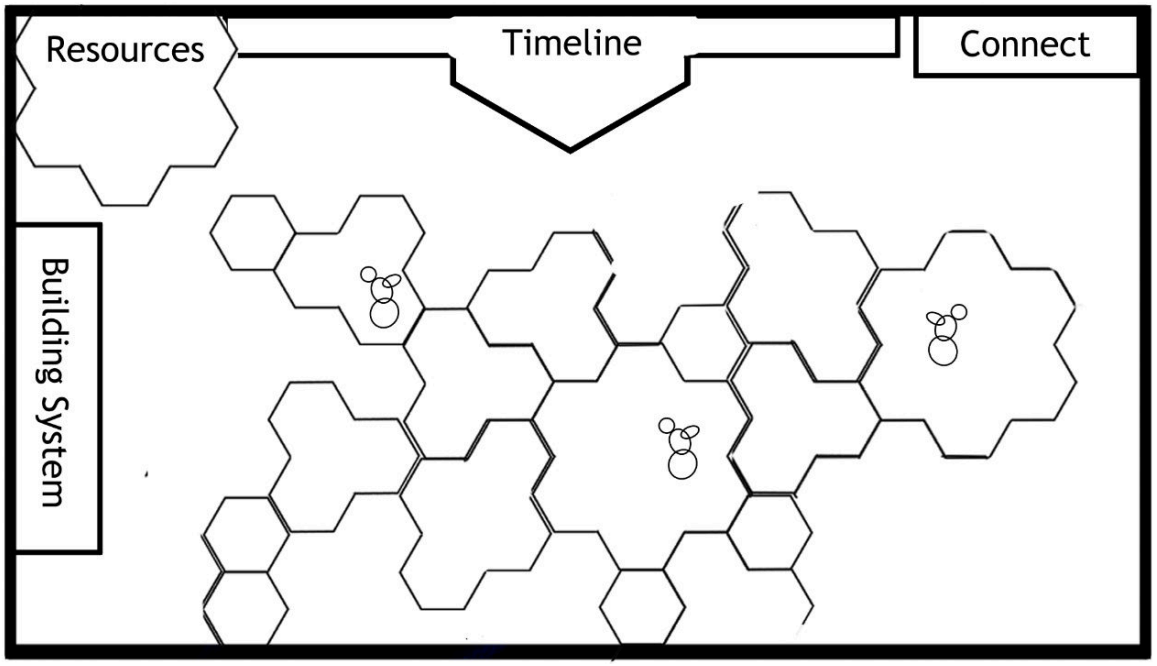
Create creates a brand new empty hive from the start, using the **New Hive Name** as its name for the save file, **Delete** deletes the selected hive, and finally **Play** loads from memory the hive

Loading Screen



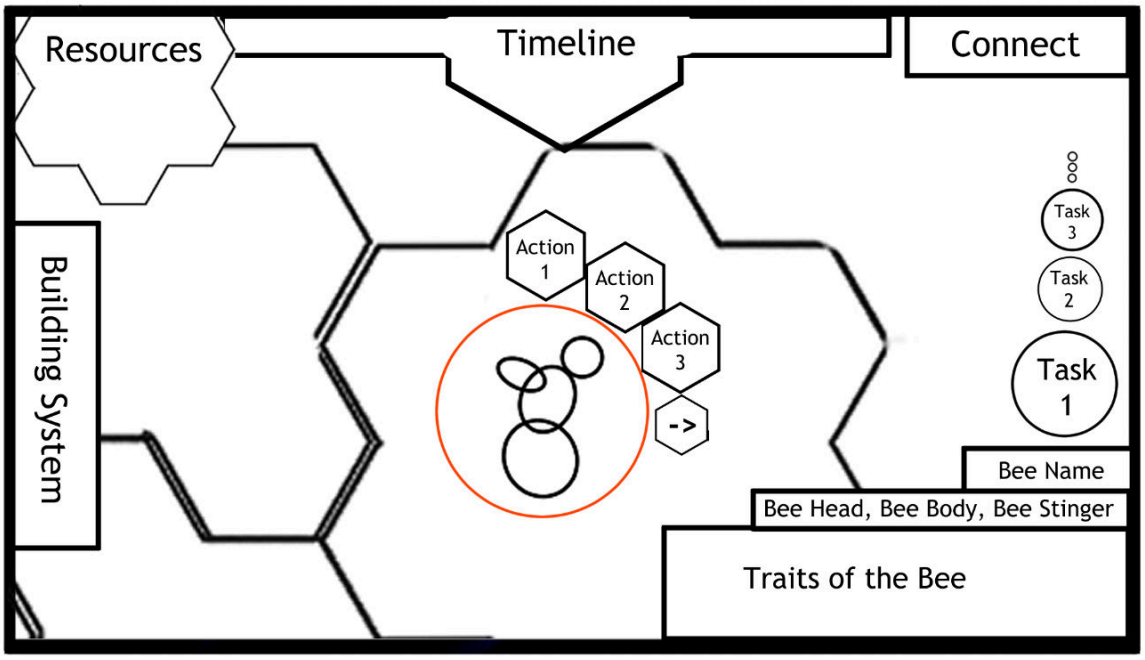
The loading screen appears once a hive has been selected and while it's being retrieved from memory. It will feature a randomly selected splash art, and a tongue and cheek randomly selected loading message.

In-Game UI



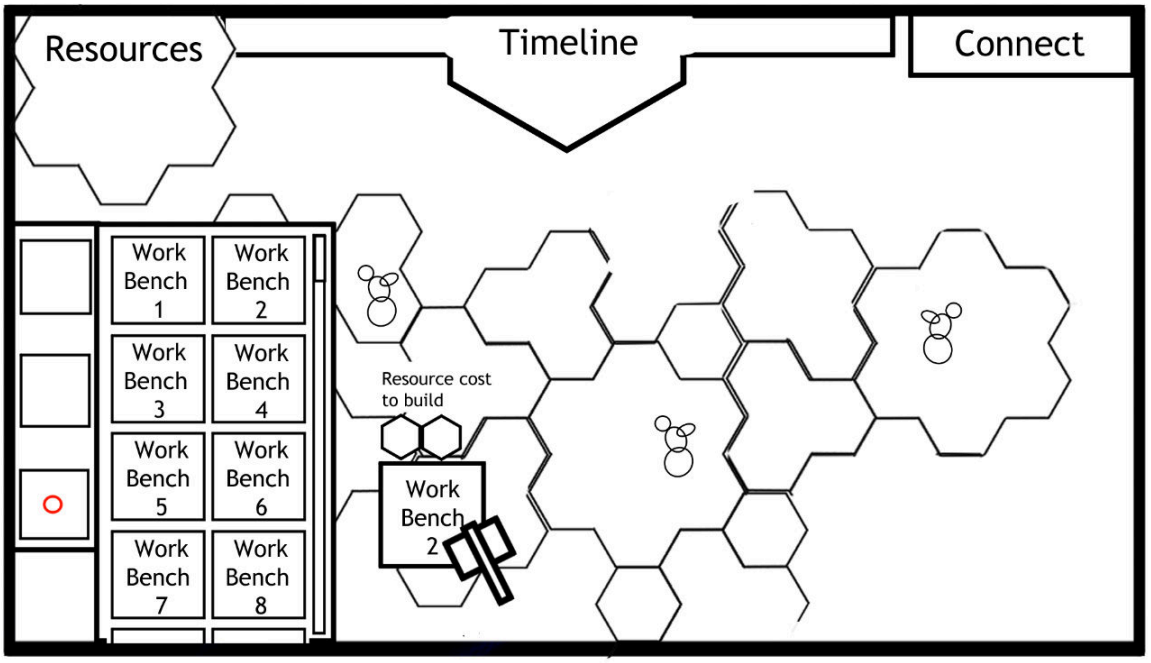
This is what the UI looks like while the player hasn't selected anything.
Resources is where the amount of each type of resources in the player's possession is shown, **Building System** is where the cell, furniture and workbench edit mode are located, the **Timeline** is where incoming event and season and the passage of time will be shown, and the **Connect** button is used to show the player's host code or to insert the host code of a friend and join them, this is also where joining request will be shown when a player tries to join the host's world.

Game UI - Bee Selected



When a player clicks on a bee, the camera zooms in and centers on them, showing in the bottom right corner all the important information about the bee, like their **Traits**, their species make-up broken into parts, and their task queue. When the bee has more than three tasks lined up, the UI will show three suspension dots above the third task; Tasks are executed from bottom to top.
Additionally, when clicking on a bee a radial menu will pop up showing all the actions and tasks the player can ask for the bee to accomplish, the bee will then add that task to its queue and perform it once it has performed its other tasks. Players can remove tasks from the bee's queue by simply clicking on them. Clicking on the right-most arrow in the radial menu will show the next three actions that are available to the bee, if any. If the bee has three or less actions available, the arrow won't show.

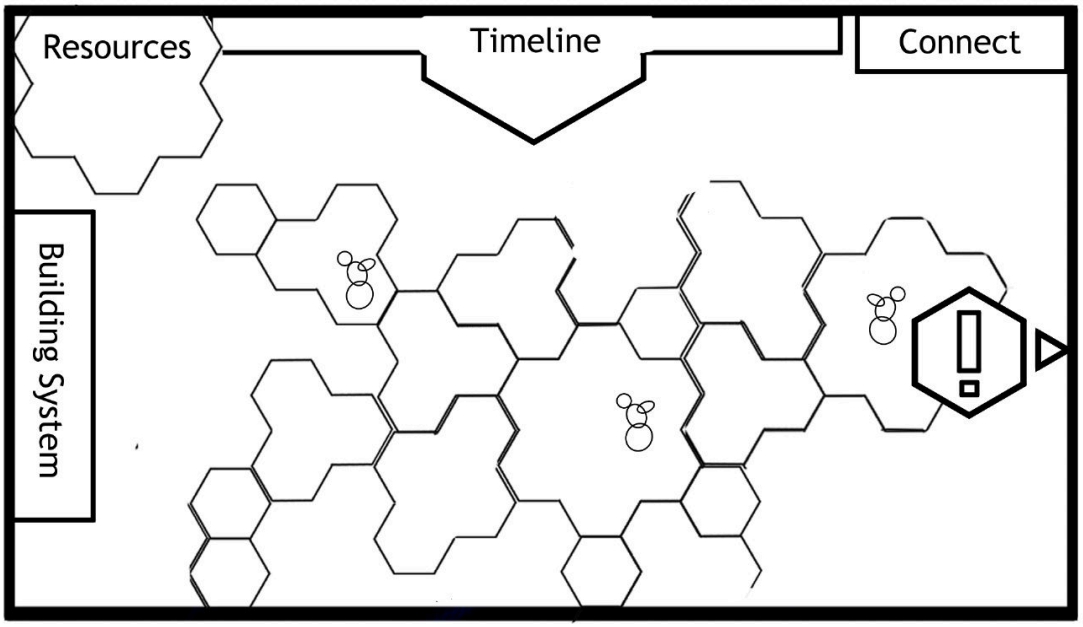
Game UI - Building



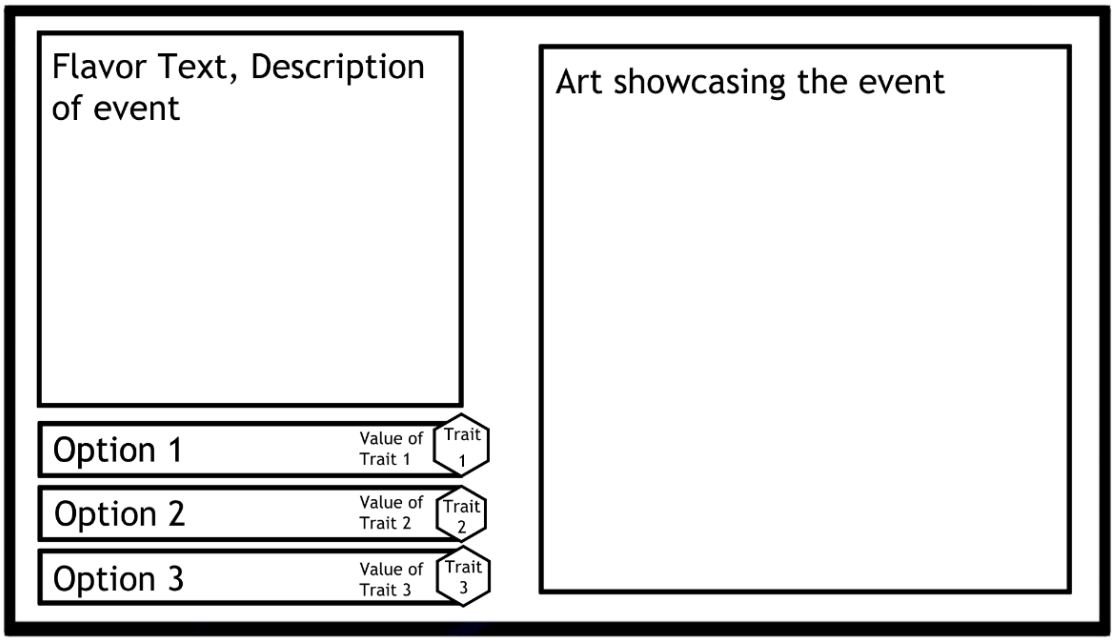
This is what it looks like when the player is trying to build something. If he clicks on the furniture button or, like in this case, on the workbench button, a sliding window appears showing all the possible options to the player.

The player can then click on one of the possible elements, and place them somewhere within the hive, while the resource cost to build the element is displayed above it. While in this mode the player's cursor turns into a **small hammer** for all players!

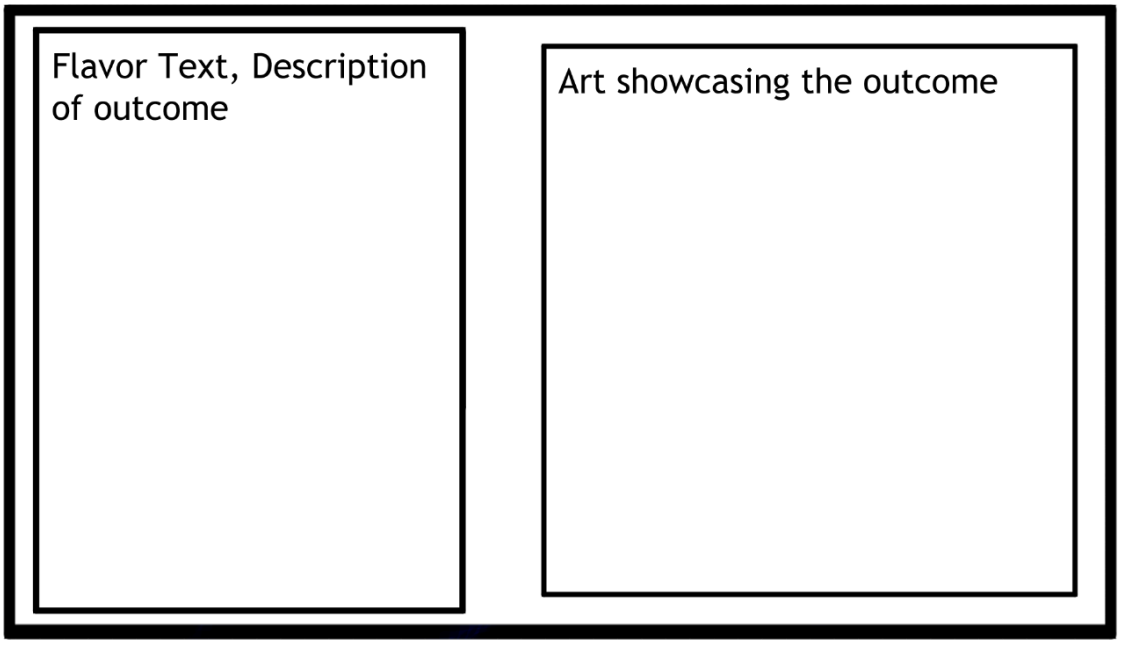
Foraging Events



When a foraging event occurs to a bee, an indicator will show up in the middle of the right-most side of the screen, the player can then click onto the indicator to see the event's interface and interact with it.



As explained in section 5.2.8, the player here will be presented with a text describing the event, art connected to the flavor text, and three options, each connected to a specific trait of the bee. Each trait will also be showcasing the value the bee has in that trait on its left.



Once the player selects an option, they are presented with the interface showcasing the outcome of their decision, which can be positive or negative. The player then returns standard In-Game UI.

5.3.4 Rules

In this section are listed the main rules of the game, organized by category.

5.3.5 Hive/World rules

- An in-game day lasts 24 minutes.
- An in-game season lasts 60 days.
- In-game time only moves forward, with the option to pause it; rewind is not allowed.
- Resources cannot assume negative values, If a certain cost cannot be paid (i.e. there are not enough resources), that action cannot be performed.
- The hive is formed by standard hexagonal units.
- Hexagonal units form rooms.
- Placement is valid only if hexagons are connected along at least one complete edge (not floating hexagons allowed).
- Hexagons cannot overlap one another.
- Furniture must only be positioned inside hexagonal units (depending on each furniture's size).

5.3.6 Player's behaviour

- The game is presented in a 2D cutaway perspective, allowing the player to move freely on the four directions (up - down - left- right).
- The player can increase the in-game time speed to a maximum of three times, which can be restored to its default value (normal speed 1x).
- It is possible to pause the game match, in order to give to the player time to think/analyse his next moves (and of course the time flow can be restored to its previous value).
- The player is not allowed to give orders to the queen bee (it acts independently).
- The game allows the player to select one or more bees at a time, in this way every bee selected will receive the same order (issuing orders doesn't have any limit, it is allowed whenever and wherever).
- Dispatch random events present three options, the player can take one option, otherwise the event duration will expire and no action will be registered (as if that event never existed).
- The player can declare cell edit mode and the game, allowing the player to modify the hive structure by building/destroying hexagons and positioning/removing furniture (can be rotated by 60° or mirrored).
- Once the player starts to draw, all hexagons traced until release, form a room.
- It is not allowed for the player to destroy the initial seven hexagons and the nursery in building mode.

5.3.7 Bee's behaviour

- Every bee has a life span expressed in days, this counter starts at the birth of the bee and decreases every in-game day, when the counter reaches zero the bee dies (life span counter may differ between species).
- When a bee receives an order from the player, it will be appended to the bee's queue.
- Every bee's actions are managed through their action queue and executed on a first-come, first-served policy.
- The bees can exit the hive only going on expeditions.
- Every bee is free to move inside the hive.
- Bees have different moods, which affect their behaviour (for behaviour effects go to Moods - Bee - Types - Game characters design).

5.3.8 Multiplayer rules

All rules from the Player's behaviour section are still valid rules.

- The maximum number of players allowed is 4.
- In multiplayer mode hives are merged by aligning them in order to maximize adjacent hexagon connections.
- Every player can give orders to all bees.
- Every player is allowed to enter building mode on the merged-hive.
- When a player builds new hexes, their ownership is determined by the initial anchor point, more precisely the first hex from which construction started (e.g., if Player 1 starts building from a hex belonging to Player 2, the new room will belong to Player 2).
- Players can only destroy what belongs to them (in building mode).
- When a player places new furniture, its ownership is determined by the hex it is placed on(e.g. if Player 1 places furniture on a hex belonging to Player 2, the furniture will belong to Player 2).
- Dispatch random events remain individual (not shared decision), and are independent from a player to another.
- Every player can share his resources by declaring a certain amount, which is transferred to the receiver.

5.3.9 Game progression

Hivemind isn't the usual game, where progression is represented by a level or a certain area of the game map. In fact this artificial life simulator game doesn't have any levels, the entire gameplay takes place inside the hive, which is day by day expanded and customized by the player's will. In order to expand the hive, the player needs resources for building new hive cells and furniture. Therefore we can say that the hive itself tracks the player's growth/progression.

But in this case, the hive is not the only indicator of game progression. There is something else which is expanding as the player manages this hive. It is the bee colony, of course !!! As the player progresses through the game, the colony evolves alongside the hive. Bees are highly valuable, as different species can be obtained , each with its own unique traits. Since different bee species have their own distinct traits, the growth of the bee colony is intertwined with the growth of the hive. This aspect is an important indicator, as the number of bees gradually increases, making the ongoing challenge of colony management progressively more demanding for the player.

Another factor to consider in the game's progression are resources. As mentioned earlier, there is a mutual progression between the hive and the bee colony, and as a final ingredient also with resources. Therefore, the overall progression of *Hivemind* can be determined by the interplay of these three aspects.

Finally, about normal games, once you arrive in a certain area very advanced you are sure that you will only proceed further. In Hivemind, anyone tells the player that once obtained the Killer bee, he will be safe forever. Unfortunately for the player there are no checkpoints here, potentially the player can lose all his bees (except the queen, which is immortal). This possible **game regression** always forces the player to weigh his choices carefully and maintain a constant challenge through the entire game.

5.4 Modes and other features

Our game is designed to allow for a seamless transition between the two main game modes, single player and co-op, without the need of an independent lobby system managing a separate scene and UI.

Single player - Host only
Single player is the foundation of all game sessions and the only access point to the multiplayer mode. To ensure a better user experience with a smoother transition from single player to multiplayer, the session is initialized as a local listen server, meaning that the host has requested a relay allocation from the Unity Relay Service and has created a lobby with both Unity Lobby Service and Steamworks API.

- **Session initialization**:
  Players select the hive they want to use for their current session from the main menu, before the session itself has been initialized. This approach allows players to confirm they chose the hive they wanted to use before joining someone else's session.

- **Local RPCs**:
  Even in single player, the game follows the same framework as in multiplayer, with ServerRPC calls. Since in single player the user is effectively the authoritative host, these calls are not sent through the network, minimizing the latency.

- **Smooth joining**:
  Once the hive has been selected and the single player session is completely initialized, the "Join by code" and "Join by Steam friend" features become active and other players are able to join the session.

Co-op mode - Merged hives
Co-op mode is an extension of the single player mode that focuses on the hive merging mechanic and the possibility of interacting with every other players' entities.

- **Join flow**:
  Transitioning to co-op requires a handshake process as described in detail in section 5.3.2.

- **Hive merging**:
  Once a client has been accepted by the host, the client sends his hive data to the host for computing the optimal position and then broadcasts to the other clients.

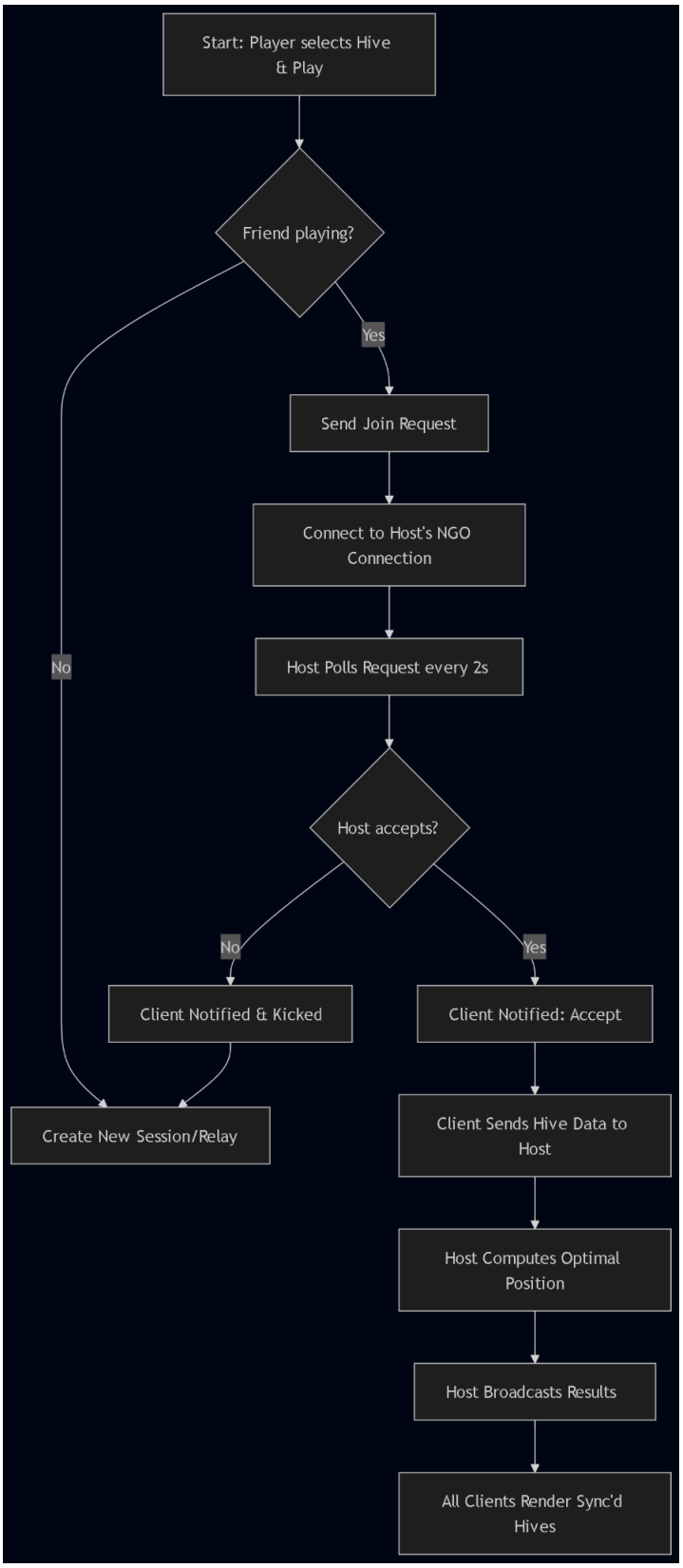
- **Shared authority**:
  Once the co-op session is fully initialized, the merged simulation becomes a fully collaborative environment, where all players can interact with all entities (bees, furniture, workbenches) of every other player. Even if all entities are shared, all actions performed by the clients need to be validated by the host to ensure that the data always remains synchronised.

- **Session persistence**:
  If a client leaves the session, their progress is saved, allowing them to resume from the same point they left on their next session, both in single player and co-op.

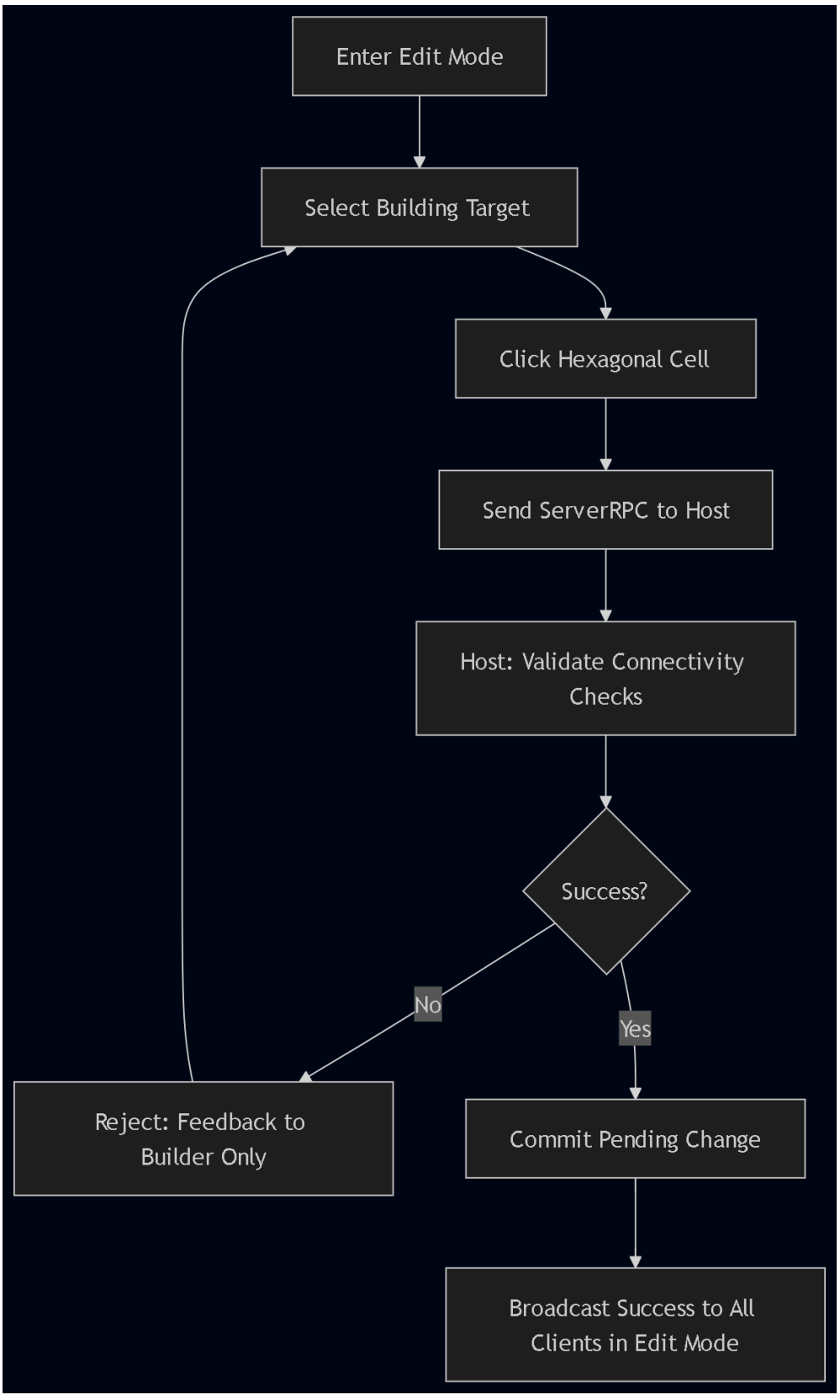
5.5 Flowcharts

5.5.1 Procedures flowcharts

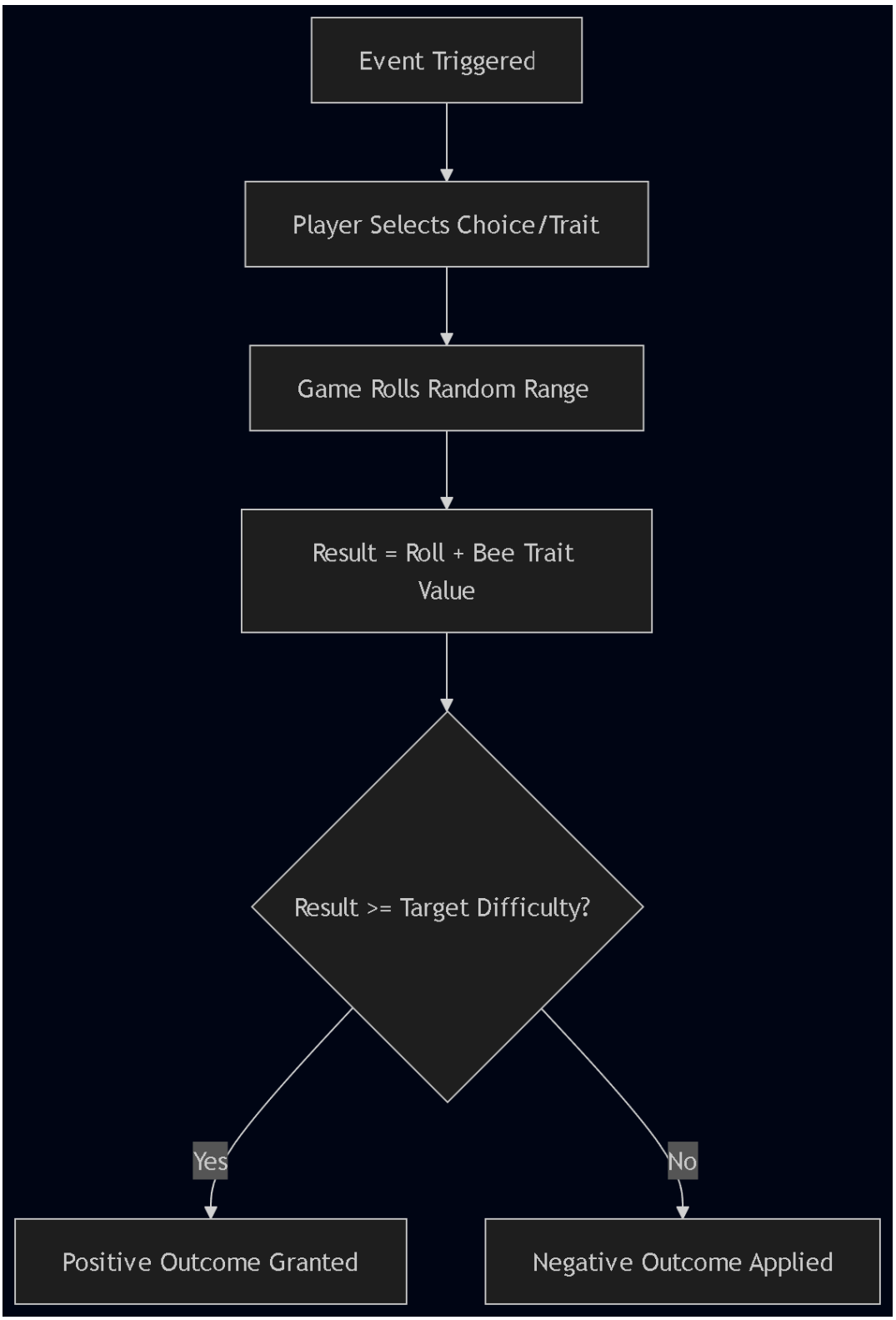
Multiplayer session connection



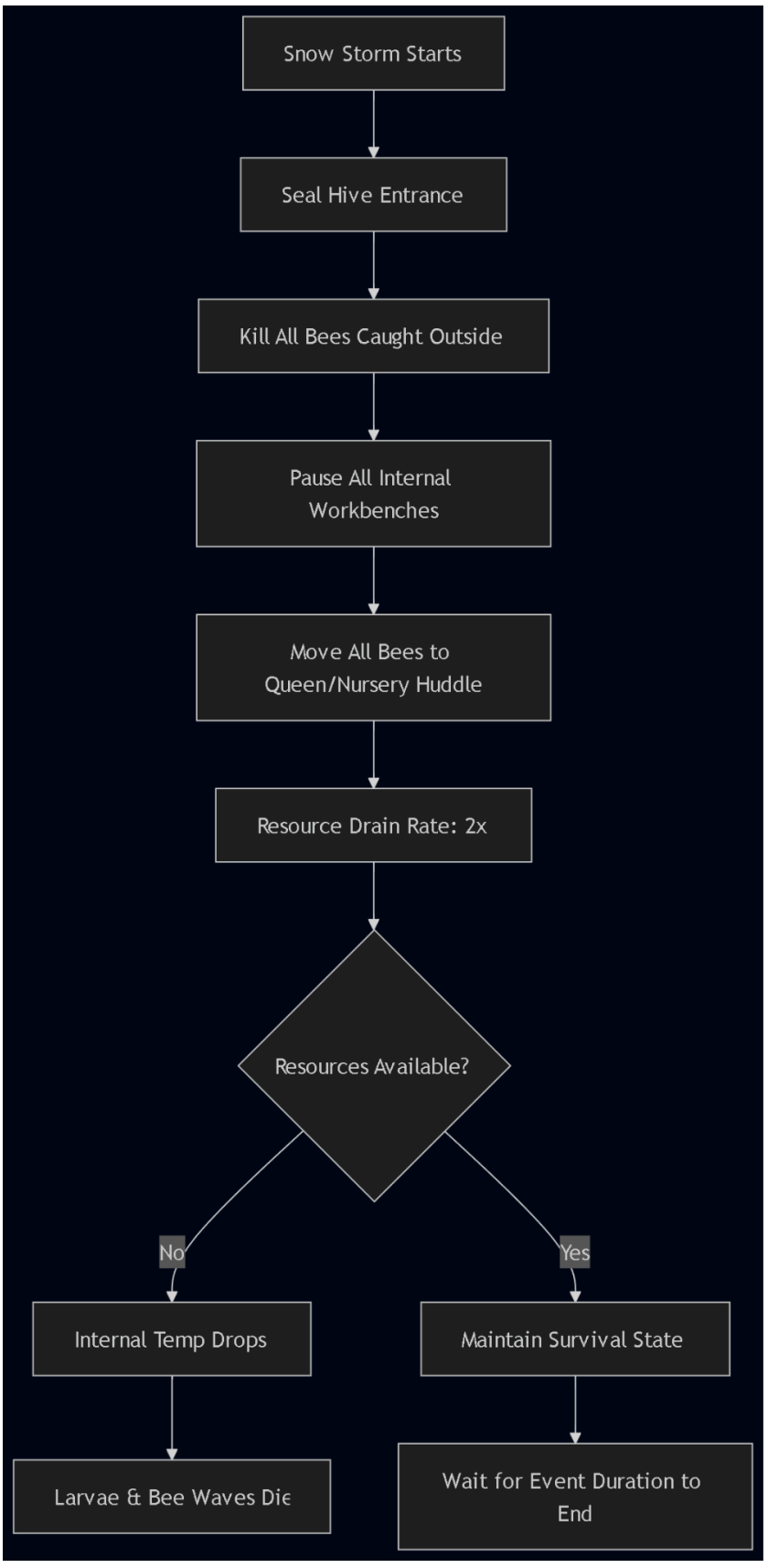
Cell edit mode



Foraging event logic

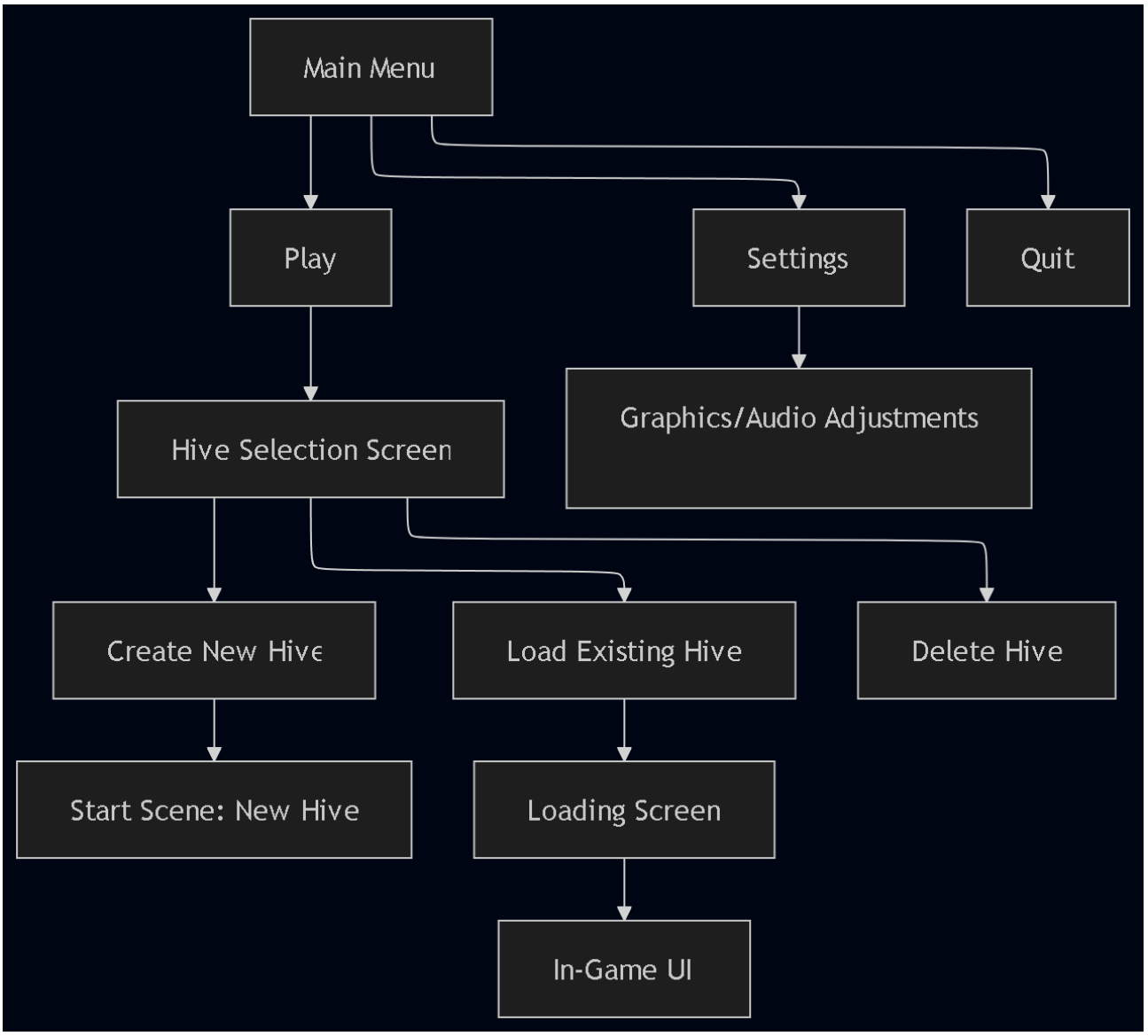


Winter snow storm

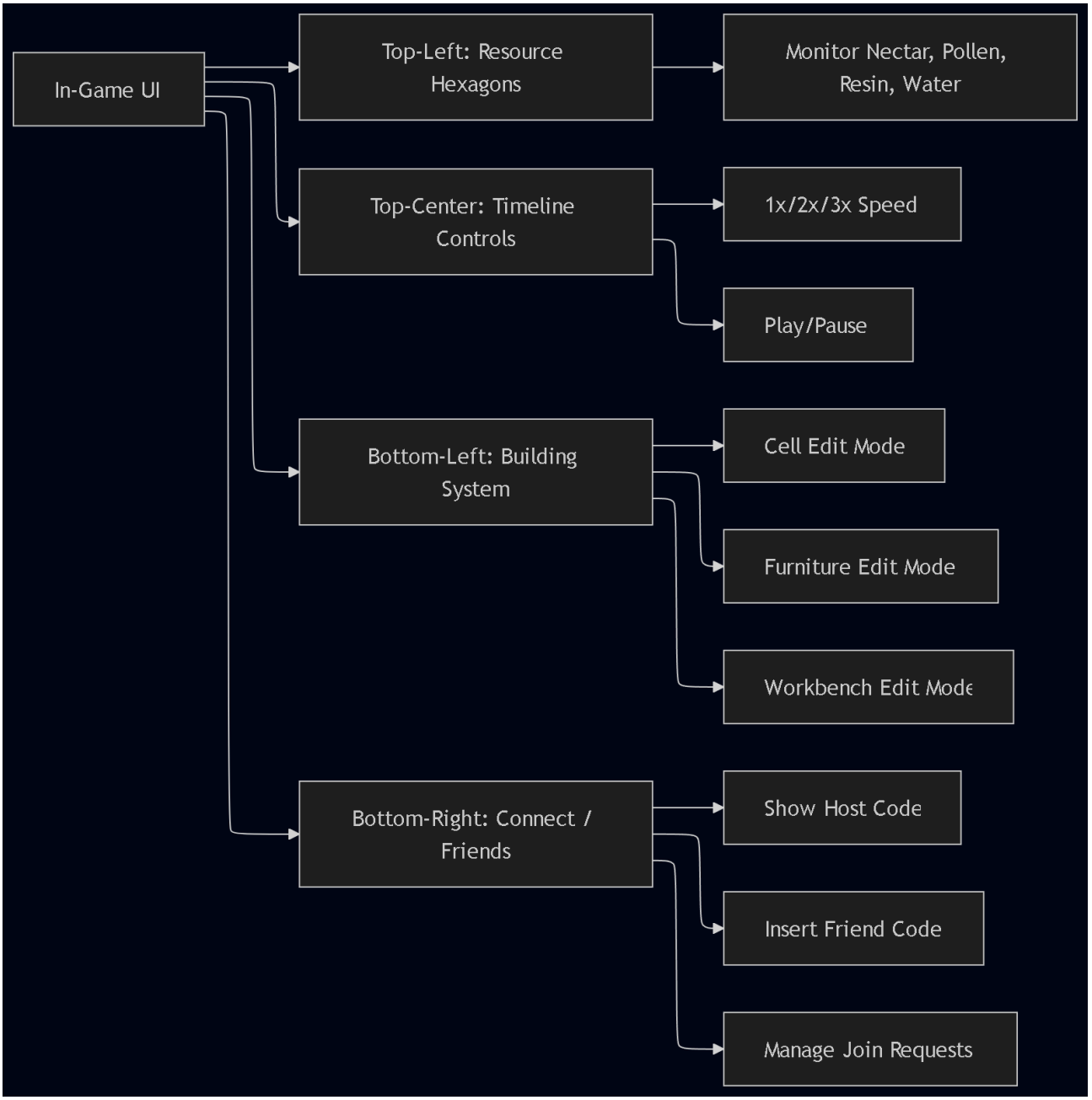


5.5.2 Interfaces flowcharts

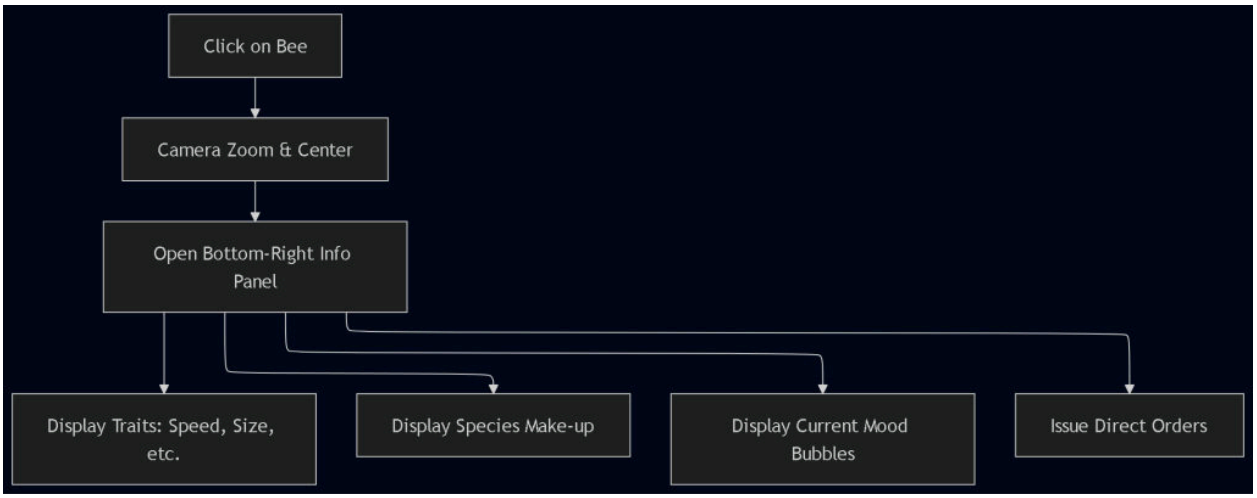
Main menu navigation



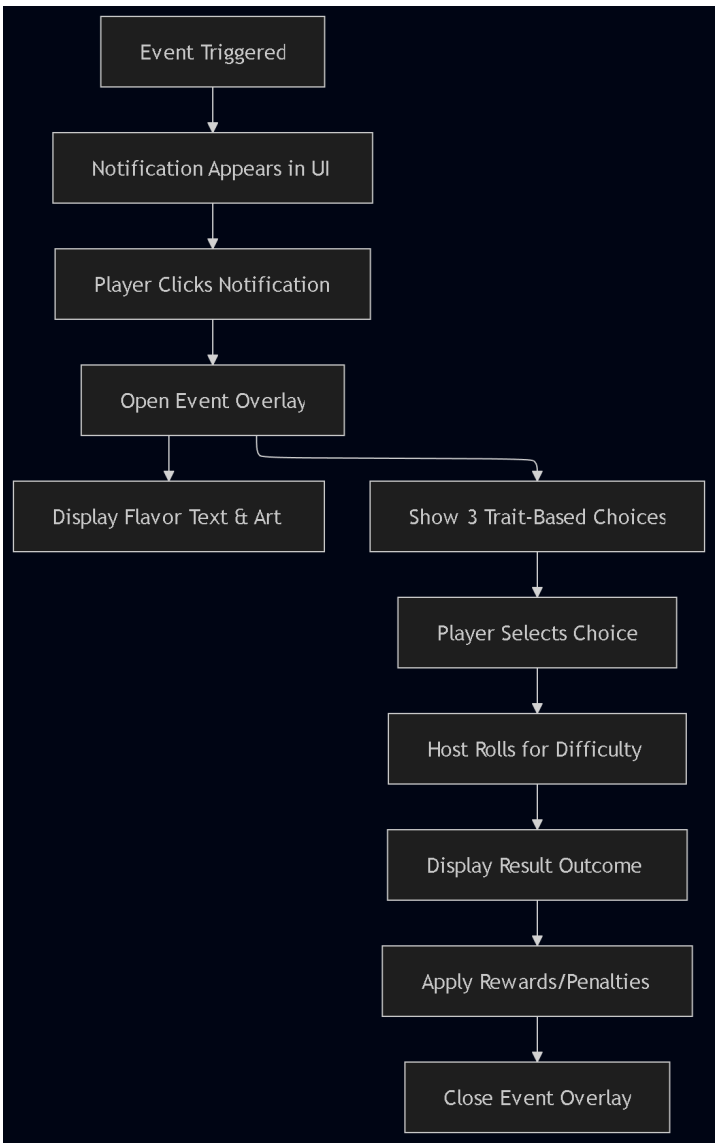
In-Game UI


In-Game UI
Top-Left: Resource Hexagons
Monitor Nectar, Pollen, Resin, Water
Top-Center: Timeline Controls
1x/2x/3x Speed
Play/Pause
Bottom-Left: Building System
Cell Edit Mode
Furniture Edit Mode
Workbench Edit Mode
Bottom-Right: Connect / Friends
Show Host Code
Insert Friend Code
Manage Join Requests

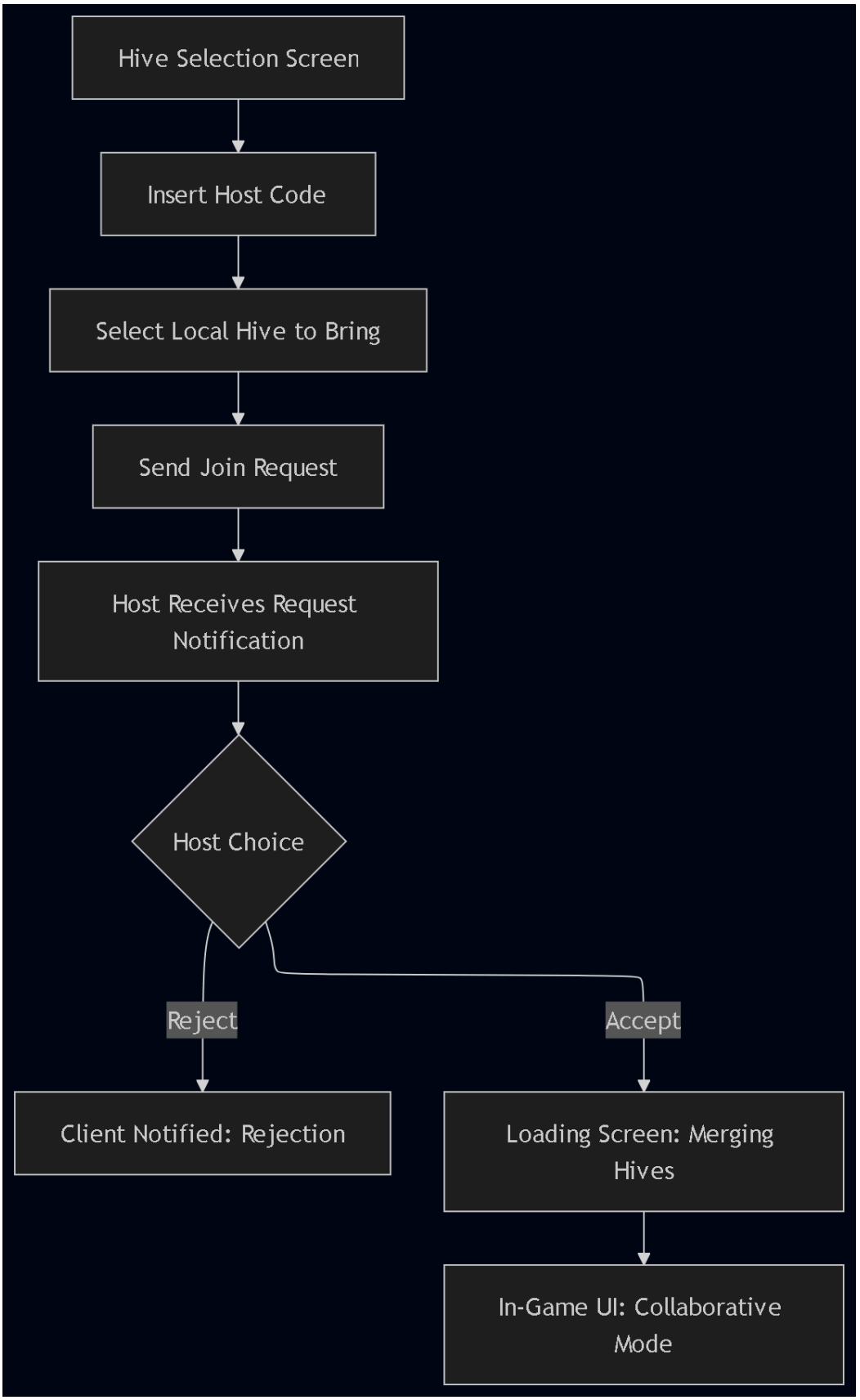
Bee details



Foraging event



Session joining

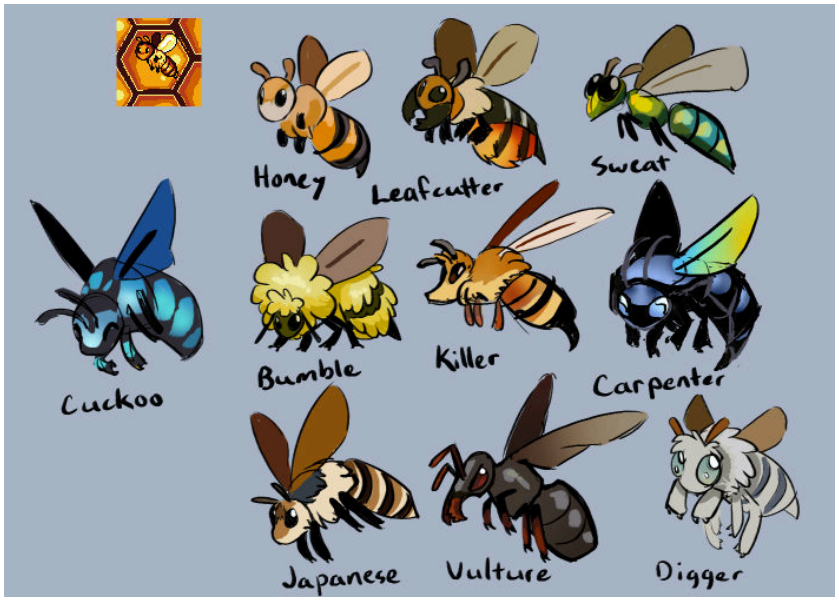


6. Game Characters

The bees the player collects, breeds, and cares for are the main characters of the hive. They represent the workforce, which allows expansion of the hive and production of important resources, while also defending the hive against the threats posed by nature and by potential invaders - the enemies they'll have to fend off.
But bees aren't meant to be just tools - to invite the player to form bonds with them and care for them, each bee is meant to look and feel different in various aspects, such as how quickly they buzz around or how often they get hungry. Every variable summed tries to subtly create a personality that the player can get attached to and recognize over time.

6.1 Characters design

When designing the bees that populate the hive, we decided to focus on two primary aspects: visual intrigue and rewarding mechanics. Having the genes of a specific species in your hive should be noticeable and rewarding, so it was necessary for the designs to differ significantly from each other in shape and color.



Each bee was conceptualized to keep its head, body, and stinger visually distinct from the others, creating unique and recognizable designs when the genetics are mixed into a new larva.

The designs are inspired by the real-life counterparts of the bees we chose, and for each we tried experimenting with shapes and colors to create greater visual variety and emphasize aspects of their biology that were most iconic, such as the green tint of the sweat bee, the size and fluffiness of the bumble bee, or the larger legs and eyes of the digger bee.



Neon Cuckoo Bee



Trigona - Carpenter Bee



Anthrophora - Digger Bee

The concepts were then adapted to the cozy pixel art style of the game, with simple palettes and thin outlines inspired by games such as **'Stardew Valley'**. The three parts of the bee are each their own sprite, and they are rigged together using an armature to create simple, playful, and most importantly repeatable animations across all bees, which create an intentional 'rag doll' movement inspired by

‘Mewgenics’. It is important that the bee’s head always stares at the player to give a greater personal connection.



Evolution of the Honeybee’s final design



Mewgenics animates several different cat ‘body-types’ and parts using the same animations and swapping out sprites to create a sense of fun surrealness, which fits the tone of the game and simplifies game development since animations don’t need to be perfect for every cat; they just need to be precise enough.



The color palette of Stardew Valley achieves visual clarity using only a few tones for every shade of color and keeping saturation minimal for a more pastel look.

6.2 Types

The Queen



Concept of the honeybee queen

The queen of the hive, inspired by the western European honeybee queen, acts as the player's avatar throughout the game. Each player has their own queen of the hive, even when they join their hives together, and are identified by the queen's regal hat.
The queen is also the source of larvae, which she will periodically produce for the player.

She cannot die from any event, and the starter room where she resides, meant to correspond with the center of the game world, has a nursery that cannot be destroyed or removed.
This means the queen acts as a failsafe for the players; should something go awry, they will always have their queen as a last resort to give them more bees and start over safely.



Concept of different crowns for the queens of different players in the same world

Bees

Bees come in many shapes! Each defines a brand new personality that the player can get to know and love. The bee's personality is defined using a system of traits that influence their behavior and their life; These traits are defined starting from a base value and then adding a random modifier for variation and uniqueness.

Traits

These are meant to flesh out the identity of a bee, a sheet of statistics that is experienced by the player through gameplay. Every trait has a range of values it can be as described in the trait range column.
Note: For all examples proposed, any number is representative and not indicative of real values. We are using a scale of floating point numbers from 0.00 to 10.00 with 0 being the lowest stat possible, 10 being the highest, and 5 being the average for that specific trait. What a '5.00' actually represents will differ greatly between any given two traits. After modifier application some traits may go above 10 or below 0, which again is purely representative of how proficient a bee is in that trait.

Trait	Description	Trait range
 Life Span	Affects for how long a bee will live for in terms of days. Without accounting for other causes of death, this represents how old a bee can get. A longer lifespan can be beneficial if a bee has useful traits but also detrimental if the player needs to rapidly change their hive. A higher **'Life Span'** means the bee will live for longer.	[2.50, 8.50]
 Hunger	Affects how much food a bee needs to eat daily. If a bee doesn't eat they'll daily amount they'll grow hungry A higher **'Hunger'** means the bee gets hungry less often.	[3.00, 7.50]
 Speed	How quickly bees move around. Affects their movement speed around the hive, how quickly they move from one place to the other and how long foraging takes. A higher **'Speed'** means the bee moves faster.	[3.75, 8.00]
 Size	Affects the visual scale of the bee within the game world, purely aesthetical. A higher **'Size'** means the bee will be larger.	[3.50, 6.50]
 Energy	Affects how long a bee needs to rest daily. If a bee doesn't rest the amount they need, for example if their sleep is interrupted, they'll grow tired. A higher **'Energy'** means the bee will need to sleep less often.	[3.75, 6.25]



Resource Gathering

Affects the amount of resources collected by the bee when returning from foraging.
A higher **'Resource Gathering'** means more resources will be collected by the bee.

[2.50, 7.00]



Work Speed

Affects how quickly a bee will perform and complete tasks. While speed affects how long it takes for a bee to get to a task, this specifies how long the actual task takes - such as making honey, building hive cells and so on. Also affects how long foraging takes.
A higher **'Work Speed'** means tasks take less time to be completed.

[4.00, 7.00]



Aggression

Affects how much the bee contributes to dispatching threats. Since enemies within the hive take a collective effort to be fended against, this trait encapsulates how much the single bee contribution is worth.
A higher **'Aggression'** means the bee's contribution is worth more in a defense event.

[3.00, 7.00]



Sweetness

Affects how likely a bee is to attempt adding their genes to a growing larva without the player input.
A higher **'Sweetness'** means the bee is more likely to interact with the larvae on their own.

[2.00, 7.00]



Luck

Affects how difficult events will be when the bee is sent out foraging.
A higher **'Luck'** means the bee will be targeted by events with a lower 'target difficulty'. As explained in section 5.2.8.

[0.00, 10.00]

Species & Mutations

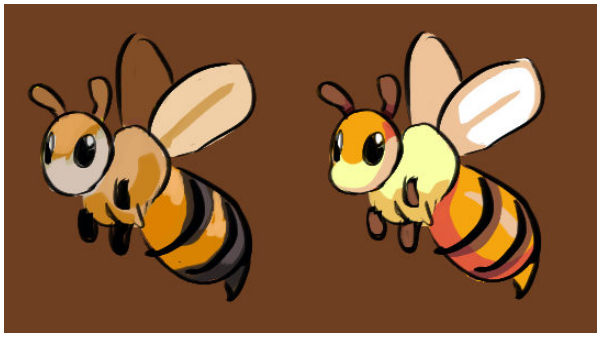
After their traits are decided, the bee's species and mutations are taken into consideration. Each of the three parts of the bee has a chance to belong to a different species - depending on which bees participated by adding their genes to the larva in the incubation period, or be mutated, meaning it'll be an entirely unique part with its own modifiers. Both of these directly affect the traits of the bee.

Each species is designed with ups and downs to have them within the hive; their usefulness is entirely dependent on what the player needs or what events are coming up that the player needs to defend against. Their life spans are designed to be short enough to allow even drastic changes to happen over short amounts of time, depending on the player's effort.

Additionally, having a variety of bees increases the social effort that invested players will have to make in the game. A player with a significant event coming up who lacks the required bees to tackle it may ask friends to join his hive and have their more adequate species pass down their genes to his larvae.

As a last remark, multiple parts of the same species on one bee do not create stacking effects; the effects are applied once for each different species in the three 'pieces' of the bee. This was done to incentivize optimization of species inheritance.
All the information and inspiration for these species was found at https://www.inaturalist.org/ *(Note: Modifiers are closely related to the available information on the bees and try to push variety rather than balance. More testing should be conducted to adjust the values.)*

Honey Bee



Apis Mellifera - the European honey bees, the traditional bee everyone knows and loves. A standard bee with **no modifiers**.

All larvae are born as honey bees if no bees of other species interact with them, meaning there is always a possibility for any given part of a new larva to be a simple honey bee

Bumble Bee

Bombus - Bumble bees, larger than most and small social, covered in hair, the most capable of bees to work in cold environments and fly to the first flowers. Its large body allows it to carry large amounts of pollen.

Modifiers:

- Reduces 'Speed' by **2.5.**
- Increases 'Size' **1.5x** times.
- Increases 'Resource Gathering' by **3.**
- The total value of 'Resource Gathering' is increased **1.5x** times.

The bumble bee is the perfect bee to pick during times of calm, where the player can leisurely stock up on resources without worrying too much about impeding dead lines. And it'll make your bee so large you'll definitely never miss it.



Carpenter Bee



Xylocopa - Carpenter bees, a big, largely hairless, tough black bee, can be nocturnal, often solitary; they build nests in wood.

Modifiers:

- Increases 'Life Span' by **2.5**.
- Increases 'Work Speed' by **3**.
- Has a chance to give back half the resources used to refine or construct something.
- Reduces 'Sweetness' by **1**.

Carpenters are those rare bees that are hard to come by, but are extremely worth it when the player does. Their solitary nature makes them hard to obtain, but once they are within the player's hive, they'll stick around for a long time.
They are extremely useful to help the player progress faster, and recuperate from disasters that affect the hive's structure directly (floods, winters)

Cuckoo Bee

A 'Cuckoo bee' is a term used for a variety of different **bee lineages** that have evolved the parasitic behavior of laying eggs in the nest of other bees and using their provisions to survive, rendering them almost entirely unable to gather resources.

Modifiers:

- Reduces 'Life Span' by **2.**
- 'Resource Gathering' is set to **2**, regardless of any other modifier.
- Increases 'Agression' by **0.5**
- 'Sweetness' is set to **10**, regardless of any other modifier.
- When genes are added to the larva for the first time, growth is sped up **2.5x** times.
- Reduces 'Luck' by **2.**

Cuckoos are meant to feel like parasites. Players can take advantage of them if they are stocked up on resources and need the manpower for production or for defending their hive, but once they are in the hive, players should be very mindful of how many they have at any given moment, or they'll be hard to get rid of.



Digger Bee



Anthrophora - Digger bees, a dwarf solitary bee, they tend to dig for their hives with their larger robust legs, all packed in a tiny body

Modifiers:

- Increases 'Hunger' by **2**.
- Increases 'Speed' by **2**.
- Moves to its tasks within the hive by burrowing, reaching places much faster. Burrowing takes **1s** independently of distance.
- Reduces 'Size' **2x** times.
- Increases 'Energy' by **0.5**
- Reduces 'Aggression' by **1.5**.
- Reduces 'Sweetness' by **0.5**.

These bees are brimming with energy and take very little from your hive, both in terms of space and resources, but their nature tends to drive them away from interacting with other bees or fighting for their home. They are not the best choice when an attacker is incoming.

Killer Bee

The **Africanized honey bee** is a hybrid of the western honey bee, produced originally by crossbreeding of the African honey bee with the European honey bee subspecies. They are typically much more defensive, react to disturbances faster, chase people farther than other varieties of honey bees, and have killed a variety of animals.

Modifiers:

- Reduces 'Life Span' by **1.5.**
- Reduces 'Hunger' by **1.5.**
- Reduces 'Energy' by **1.5.**
- Increases 'Speed' by **2.**
- Speed is doubled when fending off an attacker.
- Increases 'Aggression' by **4.**

These mean bees are the main defenders of the hive from invaders. Players might consider stocking up on them when they know they need to protect themselves, but their timing must be precise, or they'll die out too quickly.

Japanese Bee

Apis Cerana Japonica - Japanese honey bee, renowned for its great adaptability, its ability to fend off intruders by surrounding them and killing them with body heat, and its precious honey, which was a treasure in ancient times.

Modifiers:

- Reduces 'Speed' by **1**.
- Reduces 'Work Speed' by **3.**
- Increases 'Resource Gathering' by **1**.
- Increases 'Aggression' by **2.5.**
- Increases 'Luck' by **3.5.**
- Can be picked multiple times as the target of events when out foraging

The bees' rich history inspired a design that revolves around their incredible adaptability over centuries. The goal is to facilitate the acquisition of special mutations for the players, and their slower speed gives them more time to run into multiple events while foraging. The values of the trait increments were purposefully chosen so that their sum is the lucky number '7'.



Sweat Bee

Halictidae - Sweat bees, small, metallic, often iridescently coloured and less hairy bees, social to solitary, they bury their eggs with everything they need so they don't have to feed throughout the larval growth. Some are crepuscular, meaning they are active only at dusk.

Modifiers:

- Increases 'Hunger' by **2.5**.
- When eating, has a small chance to cause a larva to instantly grow.
- Reduces 'Energy' by **2.**
- Reduces 'Resource Gathering' by **1**.
- Increases 'Sweetness' by **2.**
-

Sweat bees are designed with a similar intent as the leafcutters, but with greater active input from the player. They effectively act as a 'trading system' that turns excess amounts of food into readily available bees.

Leafcutter Bee

Megachilidae - Leafcutter bees carry pollen under their body, rather than on their legs. And cut leaves to make their nest cells for larvae.

Modifiers:

- Increases 'Life Span' by **1.5.**
- Increases 'Work Speed' by **1.5.**
- Reduces 'Resource Gathering' by **3.**
- Increases 'Sweetness' by **3**.
- Larvae grow **1.04x** times faster for each Leafcutter in the hive.

Leafcutters are designed to be the heart of a growing hive. Players may pivot towards them if they need longer-lasting bees that improve and expand their home quickly. Their lower gathering trait and their bonus to larva growth specifically incentivize amassing a large number of them.



Vulture Bee

Trigona - Vulture bees, a stingless bee species that feeds on rotting meat. Some vulture bees produce a substance similar to royal jelly, which is not derived from nectar but rather from protein-rich secretions.

Modifiers:

- Reduces ‘Hunger’ by **1.**
- Increases ‘Speed’ by **1.5.**
- Increases ‘Resource Gathering’ by **2**
- Adds a chance to find special materials (meat, bones, gold…) that are worth more while foraging.
- Reduces ‘Aggression’ by **2.**
- Reduces ‘Sweetness’ by **1.5.**
- Increases ‘Luck’ by **2.**

This bee serves to incentivize exploration. Their special mechanic is a refreshing discovery the player can share with their friends, while all their other traits have been designed around creating a bee that mostly stays outside of the hive.



Mutations, unlike species, are single-part modifiers that generally give more unique modifiers to the bee or have greater bonuses (along with greater downsides). They are designed to be rare but feel extremely special and noticeable when they are found.

Mutations can also represent a risk, as they can override traits given by the species when obtained.

Angler head - The bee no longer needs to sleep.	**Tanky Body** - The bee no longer needs to eat and doesn’t use up resources when refining materials. ‘Speed’ and ‘Work Speed’ are set to 1.	**Double stinger** - The bee’s ‘Aggression’ is maxed and then doubled, while its remaining life span is halved.
Placeholder head - When the bee fertilizes an egg, its species make-up is added 5 times to the larva.	**Perfect Body** - The bee’s base traits are set to their maximum range, and then modifiers are added. The bee cannot breed, and this mutation cannot be overridden.	**Robotic stinger** - The bee’s Work Speed is set to 10.00. This bee can continue performing tasks even while in the Tired mood, but its Energy drain is doubled, and it requires 2x more Food

Infrared Head – If the bee is a soldier, it can instantly detect a Wax Moth upon its arrival in the hive. Additionally, the bee is immune to the sedative effects of the Smoker's Haze event and gains +2.0 'Aggression' during Hive invasion events.	**Lavalamp Body** - The bee is immune to Deadly chills during Snow Storms and the Frostbite debuff caused by Thermal Leakages. While inside the Hive the bee counts as a Wax Lamp furniture, preventing adjacent bees from falling into a Sad mood	**Adhesive Stinger** – The bee's Resource Gathering trait is increased by +4.00, when collecting Resin or Pollen, the bee has a 25% chance to instantly craft Glue while foraging, allowing for faster hive expansion upon its return



From left to right: Tanky Body, Placeholder Head, Angler Head, Double Stinger

Roles
The player can assign roles to the bees while they are in the hive, which will influence what the bee does when it is not being player-controlled, so the player has more control over how their hive evolves. When a bee is given a role, it is visually indicated by a specific hat on its head.

Soldier Bees - will immediately fend off against threats when spotted without the player's input.
Worker Bees - will take up any building task on their own without the player's input.
Foraging Bees - will periodically go foraging without the player's input.



These roles serve as a way for the player to have more control over how they organize their hive, which is especially helpful in the later stages of the game, where they'll want as much automation as possible.

Moods
Finally, to push even further the human aspects of the bee the player is expected to take care of, each bee will have a limited number of moods they can be in, which affect their behavior around the hive. We wanted to keep this aspect simple to avoid overwhelming the player with too much information when the number of bees grows too large. A negative mood will be indicated by a small speech bubble above the bee's head, as if they were talking to the player.
The moods we chose are limited to:

- **Happy**. The standard mood of bees. They'll behave as they are expected and buzz around cheerfully.
- **Angry**. Occurs when a bee hasn't been given a task in a long time. Builds up faster in summer, giving bees more 'Aggression', but they may start to break furniture around the hive. Players can exploit this with Soldier bees by letting them build up Aggression until a threat comes to them, as long as they are willing to let some furniture get destroyed.
- **Sad**. Occurs when the hive doesn't have enough furniture. Build up faster in winter, making bees lethargic, and they'll need to sleep more often.
- **Hungry**. Occurs when the bee hasn't eaten as often as it should. A hungry bee can still perform tasks on its own or ones that are assigned to it, but at a slower pace. If a hungry bee doesn't eat for a while they will die.
- **Tired**. Occurs when the bee hasn't slept as often as it should. A tired bee will not perform any tasks on its own and will try to rest whenever it can. A bee's rest can be interrupted by events or by player actions. If a tired bee doesn't get any prolonged rest, the bee will die.



Enemy Insects

Bees aren't the only insects wandering the world, and if players want to keep expanding their hive and explore the world, they'll have to learn how to deal with different threats. All enemies will attack through **events** - some will occur within the hive, while others present themselves only when foraging:

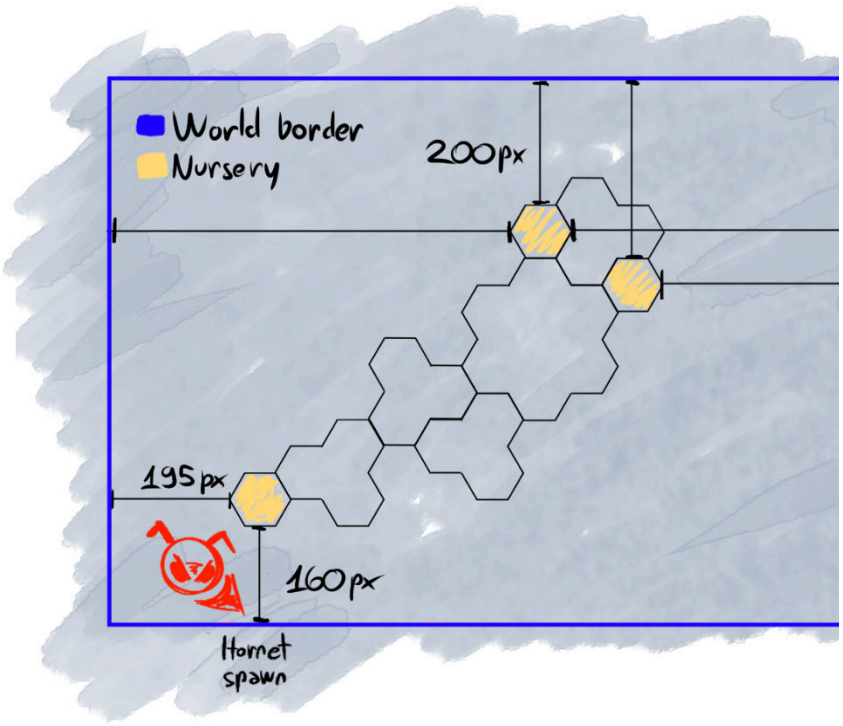
Giant Hornet - Boss Monster

Event Type: Hive event, players will be warned with time to prepare before one arrives.

Rules: This event may spawn only if the player has:

- At least **1** nursery.
- At least **10** bees.

Behavior: When the event starts, the hornet will spawn outside the observable screen, from the direction that is **closest to a larva nursery**. The hornet will then move slowly in the direction of the nursery, and once it reaches it will start striking it, killing a larva every strike.
If there are no more larvae in the nursery, it then moves to the next closest nursery.
If there are no more larvae in the hive, the hornet will leave, which marks the end of the event.



Fighting the threat: Once the Hornet spawns, all Soldier bees will immediately move towards it to attack it. The player may also select more bees to go fight the threat.
The Giant Hornet is defeated once the player builds up **2.0k aggression.** If there is more than one player, the aggression scales as such:$2000 \cdot [numOfPlayers \cdot 1.25]$.
A bee that has reached the hornet and is attacking it will add, every second, their 'Aggression' trait value to the total sum.



Gameplay Design: The hornet is designed to be a boss-like enemy in several ways. Their large 'health' pool means that players who want to take them down need to prepare in advance by adding more aggressive bees to the hive.

Their attack strategy also presents choices the players will need to answer. Is it better to have nurseries be spread apart to the edges, which guarantees one of them will be destroyed but will leave the other two safe, or is it better to cluster them all at the center, hoping their bees can dispatch the hornet before it reaches them? Players may choose to use empty nurseries as decoys, or they may attempt to empty all their nurseries before the event starts so that the hornet will immediately leave.

Visual Design:
The giant hornet is meant to be a striking terrifying creature. A loud buzzing fills the much smaller rooms of the hive, as this frightening presence approaches your most fragile units, trying to break through your toughest soldiers.



Concept art of the giant hornet

Wax Moth - Sneaky Invader

Event Type: Hive event, hidden. Players will not be warned that this event is incoming.

Rules: This event may spawn only if the player has:

- More than **0** honey

Behavior: When the event starts, the moth will spawn somewhere in the hive unbeknownst to the players. While the moth is within the hive, their inventory of honey will slowly be depleted without any discernible warning.

The Wax Moth will take **10 honey** every **5s**. If the players' honey is depleted, the moth leaves the hive, which marks the end of the event. In a multiplayer setting, the moth will take $10 \cdot [numOfPlayers]$ honey every 5s



Fighting the threat: A Wax Moth will not immediately raise the aggression of soldier bees, meaning the player will need to inspect their hive carefully until they find the intruder. Once they find them, they can select and interact with them, and soldier bees will go and attack the threat. From this point on, the players may also select more bees to go fight the threat.
The Wax Moth is defeated once the player builds up **400 aggression.** If there is more than one player, the aggression scales as such: $400 \cdot [numOfPlayers \,/\, 1.5]$.

Gameplay Design: The wasp was made to try and keep the player as engaged as possible with the game. Unlike the hornet, they present a much greater challenge for bigger hives, where players may have amassed a large number of bees and are producing more honey than the moth is stealing. Their strength comes in avoiding detection, but once a player spots it will be easy to dispatch of it.

To spot such intruders more easily, more knowledgeable players can exploit game mechanics to figure out when they are being invaded by a moth; Maybe their production of honey has suddenly slowed down, or they notice no hive events have appeared in a long time.

Visual Design:
The Wax Moth is a small, hard to see intruder, especially among a large hive of bees. It camouflages effectively from a distance to the player with its orange hues, but is quick to be noticed at a closer glance.



Concept art of the Wax Moth

Random Encounters, some examples.

Events Type: Foraging events, these encounters may occur to any bee while they are out of the Hive foraging for supplies

Gameplay Design: These events are meant to make the players really think about what type of bee they'd want to send foraging. They pose potential threats that players cannot entirely predict but can prepare against, by sending bees with better traits out into the world. Here are two examples of insects the player may encounter

Events Descriptions:

Paper Wasps - Target Difficulty: 8

Your bee weaves carefully through a dense patch of clovers, collecting nectar as the sun hangs warm overhead. Just as you prepare to move to the next cluster, a dry rustling sound comes from a nearby wooden fence post: **A paper wasp nest hangs there, disturbed by your movement**.
Several wasps peel away from it instantly, locking onto you. They begin to circle, fast and erratic, probing closer and closer.

Bull Ants - Target Difficulty: 5

Your bee settles onto a cluster of small flowers growing close to the ground, their stems tangled and hard to navigate. As you work, a faint vibration travels through the petals. Moments later, **several bull ants emerge from beneath the soil and roots, their mandibles opening wide as they detect your presence**. One climbs directly onto the stem you're perched on, while others begin to circle below, cutting off easy escape

You're weighed down, far from the hive, and very much in their territory:

- Make a break for it through the thickest flowers (**Speed**)
- Stand your ground and attempt to intimidate them (**Aggression**)
- Offer some of your nectar and quickly buzz away (**Sweetness**)

Successful Outcomes: The bee survives, additionally: choosing **Aggression** assures +25% more resources on return, and **Sweetness** lends -25% less resource on return.

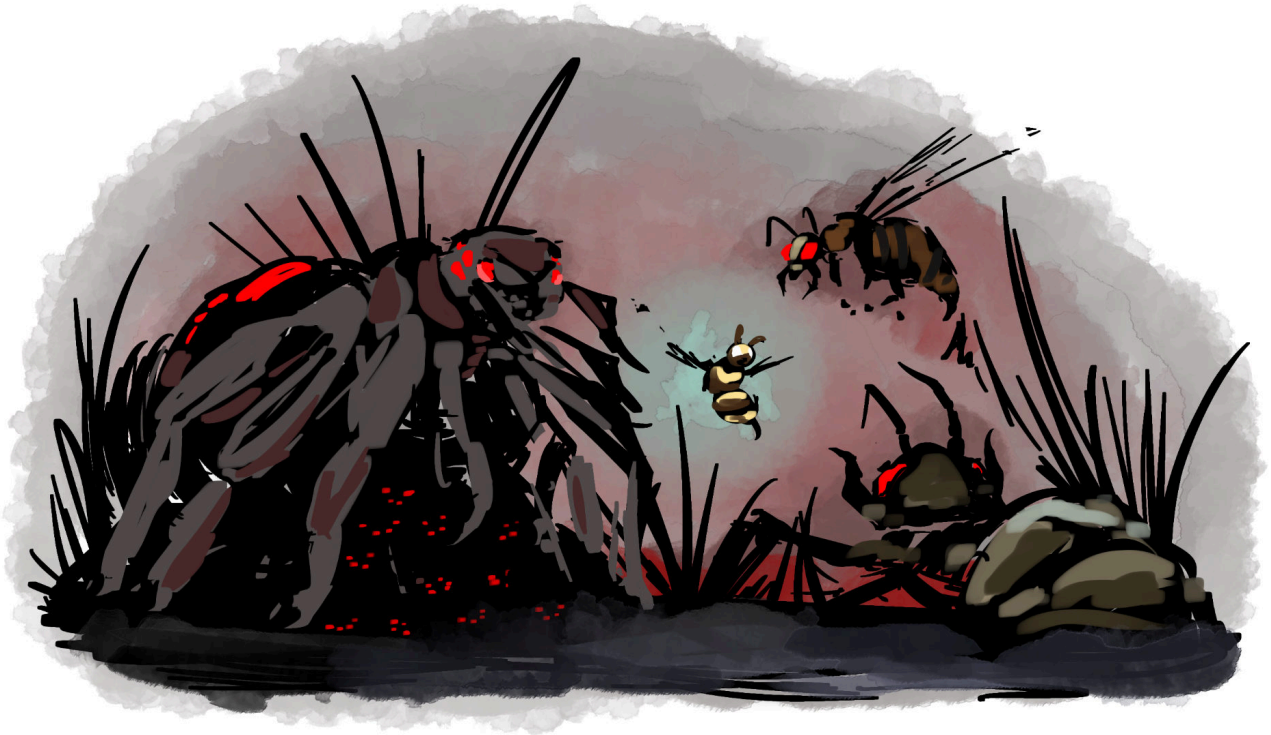
Failed Outcome: Your bee is attacked and killed by wasps. The player loses the resources the bee had gathered and the bee itself. The bee won't return to the hive.

paths.
You're cornered in tight terrain, and they're closing in:

- Turn and strike first, attempting to drive them back (**Aggression**)
- Push through their formation using your body (**Size**)
- Freeze and conserve movement, waiting for an opening (**Energy**)

Successful Outcome: The bee escapes successfully and gains **+1** to the trait the player chose.

Failed Outcome: Your bee returns to the hive with no resources. Additionally, the bee loses **-1** in the trait the player chose.

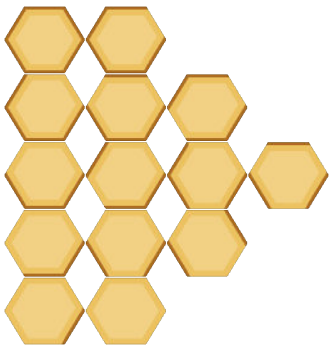


7. The Game World

In Hivemind, players get to decide what their world will look like! All within the warm confines of their beehive.
When they create a new world, they'll start with a simple hive made of 7 hexes, a nursery, and their queen bee. From there, the player can quickly expand the shape of the inside of their hive, however they wish, using the building feature (As detailed in Ch. 3). Creativity and personalization are at the core of the housing system of the game.

7.1 Build Your World, Hex by Hex

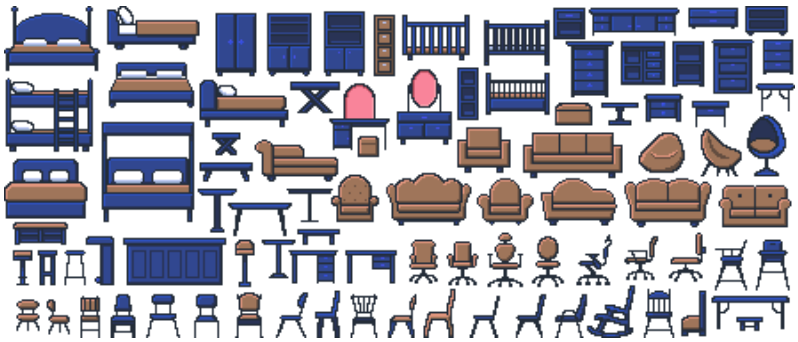




The game uses a minimal amount of 14 hexagonal tiles to let the player paint their home to their liking. Tiles belonging to the same room will interconnect smoothly with one another according to their adjacent tiles.
The golden color of the tiles and the deeper brown of the empty available space recall the inside of a real beehive made within a hollow tree trunk, where they usually reside. It serves as a good neutral background for the colorful bees to really shine as they traverse their home.

The choice of hexagonal tiles and grids was also inspired by real hives and the shape of honeycombs across media, but we preferred to use a flat top for the hexes to maintain horizontal 'floors' and 'ceilings'.

Besides just expanding, players will also need to build working stations for their bees, for honey and food production, or to build more nurseries for their larvae. Additionally, they can indulge their bees by building fun furniture for them, making their hives feel more personal and the bees happier in their home.

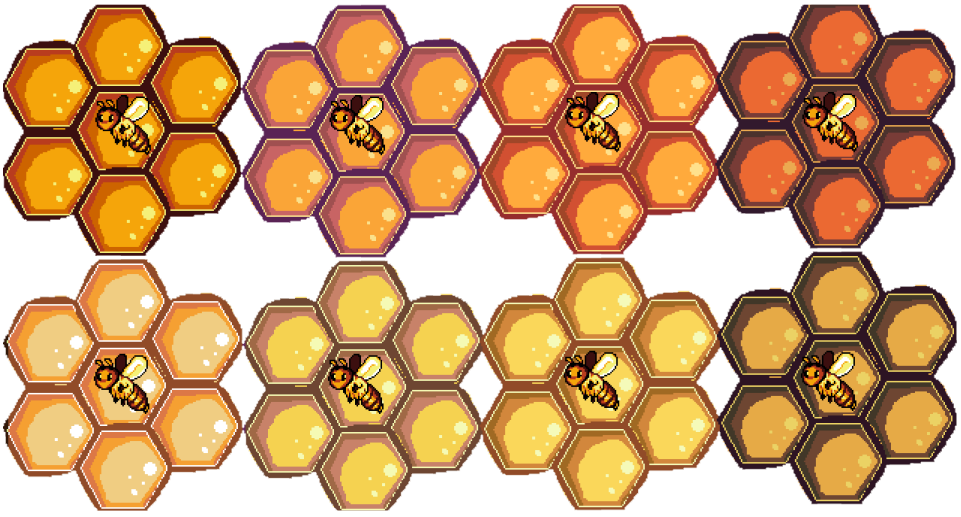




The game takes a surreal and light approach to the world with the goal of making bees feel more humanoid than insectoid. Bees collaborate with different species, they can feel happy or sad, they prepare honey in giant furnaces, and they build couches and lamps for themselves. The experience is aimed at making players feel like they are building a home more than a structure, and one that is full of warmth and chaos.



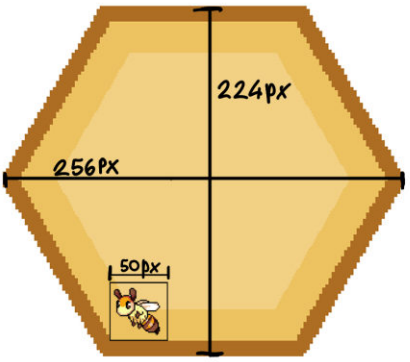
When other players join, the world grows even fuller, as the hives the players made on their own join as one into an even bigger home. It's an invitation for players to come together and see how their hives function as one, trade bee species, mutations, and resources with each other. From there, players keep expanding their colorful homes.



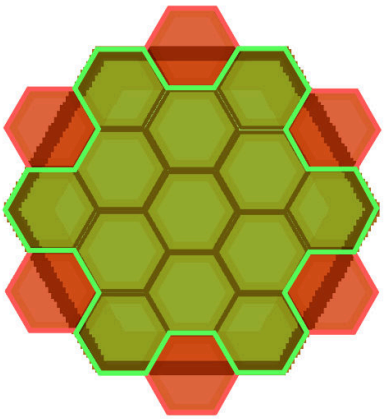
Possible color options for different players' hives.

7.2 Life Inside the Hive

The world exists on top of two grids, one controls the placement of the hive tiles, while the second is a subgrid which controls the placement of work stations and furniture. The subgrid is 1/4th of the bigger hive grid. Each element of the grid is a hexagon, meaning that all sprites need to carry the 0.86:1 proportions of perfect hexagons with flat tops. The bees' sprites are 50px, so we made the smallest elements of the grid 64px, and the larger hive tiles 4 times as large, 256px per single tile. Taking into account the proportions of a regular hexagon, hive tiles are 256x224px while the smallest subgrid element is contained within a 64x56px sprite.



This means, given a singular hive tile, only 13 elements of the subgrid can fit on top of the main grid tile. If the tile is fully surrounded, that means the player can also use the hex in the subgrid that lies between two tiles, meaning a tile can at most house 19 single hex furniture pieces.



This detail becomes important when considering the elements that can be placed within the hive, be they workstations or furniture. These are designed around the limitations of a hex grid, keeping the size of the sprite and how many units of grid it takes up closely related to one another.



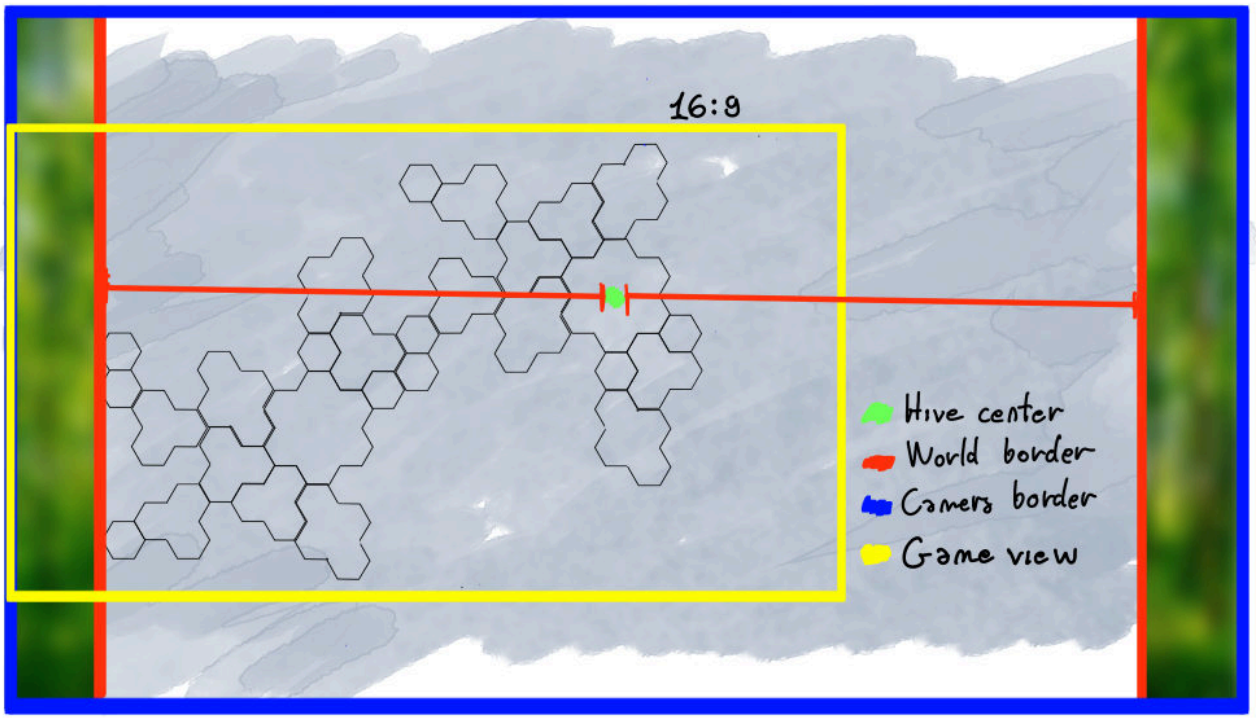
Players will, of course, have the liberty to rotate and mirror their furniture, allowing it to be placed anywhere in the hive, and against any surface, greatly improving the possibilities of customization despite the limited nature of a 2D environment.

7.3 The Outside World

The sense of safety and warmth within the hive opposes itself to the unknown dangers of the world outside, of which the players will only see glimpses when they expand their hive or when they send their bees foraging. Events, as detailed in Ch. 3, are how the world outside calls for the player to interact with it.

World Bounds

As explained, conceptually, the hive is located within a hollow tree trunk. While the game could allow the player to keep expanding in all directions, we wanted to preserve that thematic consistency by placing bounds to the left and right, spanning the width of the tree.
This decision meant that hives over time would develop more vertically, but it also gives us a chance to showcase a glimpse of the outside environment when the border is reached, a faded image of the grand world that lies beyond your humble home.



The **Game view** is what the player is actually seeing on the screen. The game uses the standard 16:9 ratio, and the player can zoom in and out on their hive to look at a singular bee or at the totality of their home.
The **Camera border** is the maximum distance the game view can be; this distance shows the entire width of the trunk of three and a glimpse of the outside world. The camera view cannot move past these borders. Left and right are static borders that depend on the size of the 'trunk', while top and bottom are dynamically adjusted depending on how much the hive has expanded vertically.
The **World border** represents the left and right-most limits for the players' building. They are static and equidistant from the **hive center**. As mentioned, players can expand freely vertically; the horizontal limitations serve more as a device for visual context and intrigue.

Seasons

The passage of time in the game is most evident with the alternating of seasons, all bringing different challenges to the game the players need to answer.
The changing of the seasons does not go unnoticed, the blurred edges of the world outside the players home change drastically, and with them also the hues on screen, and the tunes of the background music.

Spring - Easy

A soft, golden-pink light breathes life into your home. The air feels lighter, alive with the distant hum of wings and the gentle sway of fresh blossoms beyond the hive. Colors return in quiet bursts, petals unfurl, grass stretches upward, and the world outside seems to invite exploration once more. Things could not be better.

During spring, the player's bees are brimming with energy, requiring less rest and showing increased efficiency in their tasks. Foraging is rewarding; nature has plenty of basic resources to offer you and all other insects, and events are of mild difficulty. The season is a break from the oppressiveness of winter, a period during which the game feels easier and more relaxing. Take your time, smell the roses, and enjoy the blooming season.





Summer - Medium

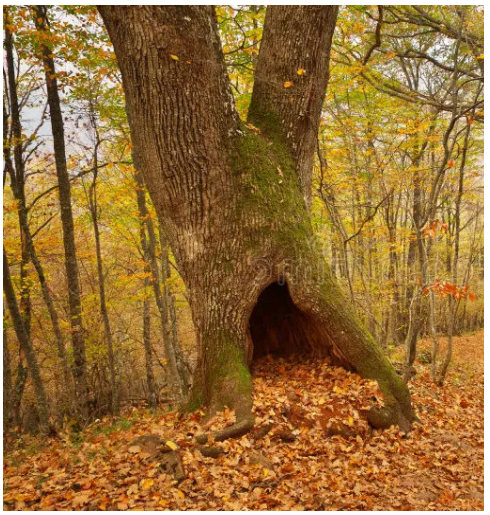
A thick, dry heat settles over your home, pressing in from the outside world. The fields are crowded with restless wings, every creature drawn toward the same riches. Colors grow intense, flowers bursting open in abundance, their scent almost heavy in the air. There is no quiet, only constant motion and competition. The world feels alive, but not welcoming. Is the wealth of the season worth the struggle it demands?

During summer, the player's bees benefit from long days and abundant flowers, making foraging more rewarding. However, they are far from alone. Other insects grow more active and aggressive around resources, leading to frequent encounters and contested gathering spots. While the colony can grow quickly, constant pressure makes expansion riskier. The season is meant to feel rich but tense, rewarding bold play while testing the player's ability to handle conflict and sustain their hive.

Autumn - Medium

A muted, amber leaf gently drifts into your home as the world begins to waver. The air shifts unpredictably, one moment calm, the next stirred by sudden wind. The once vibrant fields start to thin, their colors fading into warmer, duller tones. Movement outside grows uneven, some creatures rushing to gather what remains, others already gone. Will you gather what's left while you can, or prepare for what is coming?

During autumn, the player's bees face increasingly unreliable conditions. Foraging remains viable, but resources become less consistent as the weather interrupts activity. Events become extremely frequent for bees outside, making foraging more of a necessary risk with winter coming so close. The season is meant to feel transitional and uneasy, encouraging players to adapt to changing conditions while securing enough to endure the coming hardship.





Winter - Hard

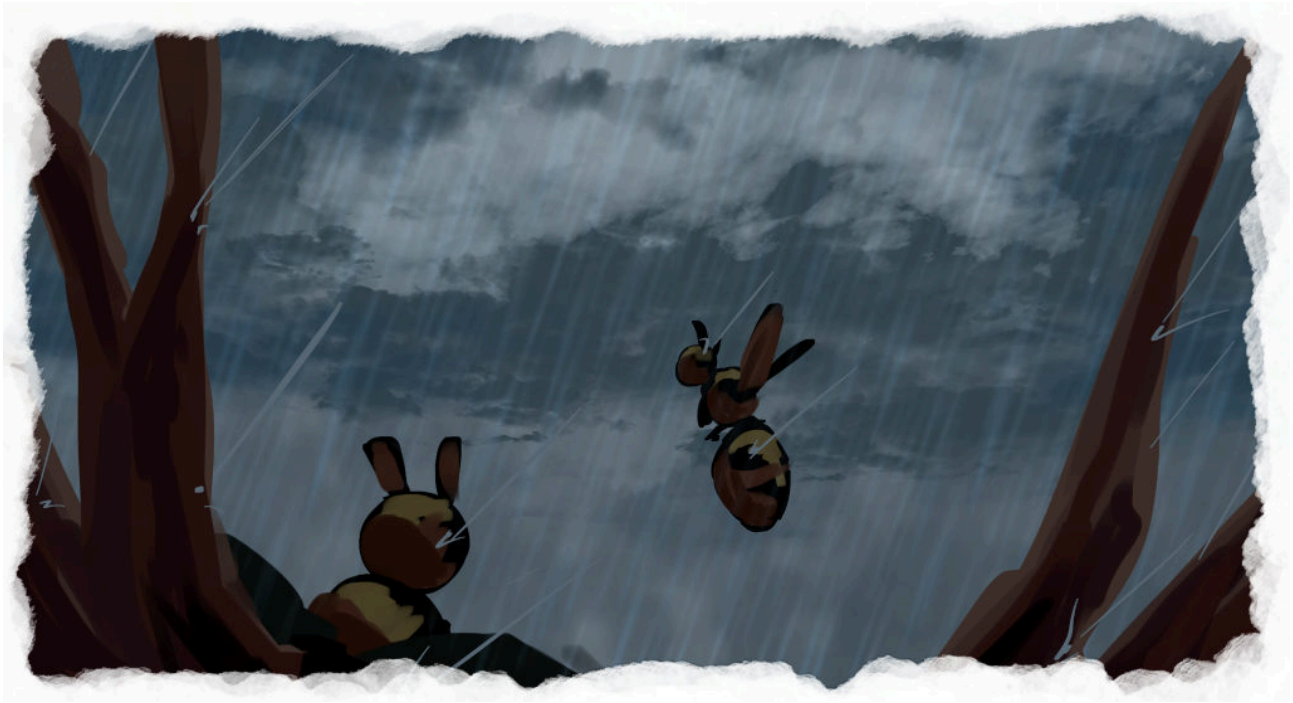
A cold, greyish-blue tint of death washes over your home. The hive grows quiet enough to hear the tune of branches rustled by the wind outside. Insects grow tired, with very little reason to go out and explore the world. Few, if any, resources remain to gather outside - Is this an opportunity for bees to venture where other insects won't? Or should they remain home and fortify against the deadly weather?

During winter, the player's bees will grow more lethargic. They'll require to sleep more often, and be generally slower to complete their task if their home isn't stimulating enough. Foraging will bring fewer resources, but other insects will not show up. The season is meant to be a challenge against nature itself, giving a sense of loneliness to the player; they can only overcome by keeping their hive compact and safe, and maybe inviting a friend or three.

7.4 Weather Events

Weather conditions are **Hive** events that depend on the game's current season. Similarly to invasions, detailed in Ch. 6, they propose different challenges the player will have to learn to face and resolve.

Can players react fast enough to prevent the winter snowstorm from snowing in their hive and destroying some of their rooms? Can they slow down the autumn flood from bloating and breaking all their beautiful furniture? Can they avoid losing resources to the unforgiving summer heat wave? Each weather condition presents a way seasons can interact with the player, even within the safety of their home.



The goal is that by combining all these threats of different natures and solutions, the players learn to be adaptable - whether that is by breeding and maintaining a hive that is ready for any situation, or finding ways to expedite new generations of their colonies to respond to every threat as it comes.

7.4.1 Spring

Pollen Frenzy:

- **Event description**:
  As the spring flora reaches its peak, a pollen frenzy envelops the hive and all its surroundings. For a short period of time, the environment is filled with pollen and other nutrients. All foraging expeditions and workbenches receive a significant production boost, while bees' capabilities and morale are improved.

- **Mechanics**:
  Global production multiplier: A 1.5X multiplier is applied to all activities producing resources, both workbenches and expeditions.
  Traits buffs: All bees within the hive receive a 25% boost to movement speed, energy and work speed.
  Timer-based event: The event duration is controlled by the global timeline, usually it lasts from 2 to 4 in-game hours, which, without any time acceleration corresponds to 2 to 4 minutes in real life.

- **Challenge**:
  While this event is primarily designed to help players bring the simulation ahead in the growth curve, the "Pollen Frenzy" real challenge is not about survivability, but rather about production

optimization and general hive management, through important and timely decisions. Failure in doing so will not result in any negative effect or debuff, it is just a lost opportunity.

Smoker's Haze:

- **Event description**:
  A nearby apiculturist has started his routine of hive inspection, using white smoke to tranquilize bees. Even though the smoke is not lethal, its sedative properties affect the efficiency of the hive. Because of the smoke some bees have been intoxicated and need to breathe some fresh air.

- **Mechanics**:
  Sedation debuff: Some bees have been particularly affected by the smoke. Targeted bees suffer from a 40% movement and work speed reduction.
  Air refresh: Bees that have been intoxicated need to be sent out foraging for an "air refresh", cleaning the debuff.
  Timer-based event: The sedation debuff is not applied only at the beginning of the event but for the whole duration of the event bees are constantly targeted by the smoke. The event can have a duration of up to 2 hours of in-game time, which corresponds to 2 real life minutes without time accelerations.

- **Challenge**:
  For the whole duration of the event players need to stay alert and check their bees to see if they have been debuffed and constantly send intoxicated bees out foraging to trigger the air refresh.

7.4.2 Summer

Giant Hornet Attack:
The event has already been explained in detail in section 6 of this document.

Heat Wave:

- **Event description**:
  The peak of summer heat brings a relentless heatwave, making the environment harsher to survive in, with cells spontaneously combusting and the risk of fire spreading to nearby cells.

- **Mechanics**:
  Spontaneous combustion: During the event random cells catch fire, with the risk of adjacent cells to also catch fire and cause an exponential disaster. Cells burn for a certain period of time, during which they can spread the fire. Once this time period has elapsed the cell is marked as burned, destroying furniture and workbenches that were inside and making a spot on which structures cannot be placed until repaired.
  Fire extinguish: Bees have to go to the nearest water source and use it to put out the fire.

- **Challenge**:
  The challenge requires the player to be aware of its resources at all times and also incentivise having bees with a high movement speed, since they can take some water and extinguish the fire faster than slower bees.

7.4.3 Autumn

Flood:

- **Event description**:
  As the tree's leaves color suddenly turns to a honey-like color, the weather is not always so calm during this season, as heavy rainfalls can damage the hive. These rainfalls may cause cracks in the hive's walls and cause some water to slip in, threatening to dampen the hive and posing a threat to precious larvae.

- **Mechanics**:
  Cracks spawning: Around a random number of cells in the outer layer of the hive enter the leaking state, causing water to slip inside.
  Dampened cells: Cells in the lower side of the hive become dampened, making bees who are

there move 30% slower and die if they stay for too long in a dampened cell. Also workbenches that are inside dampened cells completely stop working.
Automatic water drain: The water that causes cells to be damped, slowly evaporates on its own. Another possibility is to send bees to retrieve water and store it in the intended storage area for water.

- **Challenge**:
  The player must search for the biggest and fastest bees, so that fast bees can reach the cracks site and fill the hole as fast as possible, while bigger bees have an easier time filling the gaps. This event requires bees with higher work and movement speed to collect water from dampened cells. Overall this event incentivises the player to have a wide range of bees with different traits.

Wax Moth Invasion:
The event has already been explained in detail in section 6 of this document.

7.4.4 Winter

Snow Storm:

- **Event description**:
  The environment outside is covered in a white mantle, but nature is not as clement as it seems. The rigid temperature outside causes the entrance to the hive to freeze shut. All bees that were out foraging die because they cannot come back into the hive, and the situation inside is not much better. Everything comes to a halt, bees stop working and they group around the queen, trying to keep the temperature as steady as possible.

- **Mechanics**:
  Sealed entrance: Bees cannot enter nor exit from the hive, meaning no expeditions can be sent during the event period and bees that ended up outside at the beginning of the event die.
  Deadly chills: All workbenches are automatically paused upon the event beginning, bees abandon their tasks to group together near the queen.
  Resource drain: During this event no resources can be produced and bees constantly consume resources.

- **Challenge**:
  The core concept of this event is to prepare before the actual event starts. Unlike other events in this case there is nothing to fix, it is a simple but harsh endurance test. This event requires the player to accurately estimate if his resources are enough to survive in advance. If resources are low, must manually select bees to exile. Bees in the exiled state will remain inside the hive (since they cannot exit), but they will not consume any resources. This event underlines the importance of hunger and life span traits.

Thermal Leakage:

- **Event description**:
  As the temperature outside drops below zero, the hive's resin structure becomes brittle and with time it starts to crack. Unlike the flood event, this isn't an intrusion of water or snow, but a dangerous loss of heat. In order to save the hive from freezing the player must choose some bees to sacrifice themselves and seal the crack with their own body.

- **Mechanics**:
  Frostbite aura: Every active crack generates an area around which all structures and bees suffer the frostbite effect. Frostbite reduces bees movement by 75% and drains health at a steady rate.
  Sacrifice command: Player selects a group of bees to sacrifice and seal the crack.
  Permanent sealing: Each bee sent to the crack adds to a "Structural plug" meter. Once the meter is full the bees are integrated into the wall and the crack is considered sealed, removing the frostbite debuff around. The amount of meter filled by a single bee is proportional to its size and life span traits.

- **Challenge**:
  The core logic of this event is to evaluate losses, since the player is forced to sacrifice some bees.

For this event, the player has to decide whether to look for more weak “expendable” bees or fewer but chunkier bees to send to the cracks. Also if multiple cracks appear the player needs to quickly think about the priority of which crack to seal first.

8. Media List

This section details all the media assets required to develop **Hivemind**, including interface elements, environments, character sprites, animations, and the audio landscape. The game's aesthetic is characterized by a **cozy pixel art style** with simple palettes and thin outlines.

8.1 Interface Assets (UI)

The user interface is centered around a frame UI that borders the internal view of the hive, allowing the player to oversee colony dynamics and issue orders.

- **Main Menu:** Buttons for New Game, Continue, Settings, and Exit.



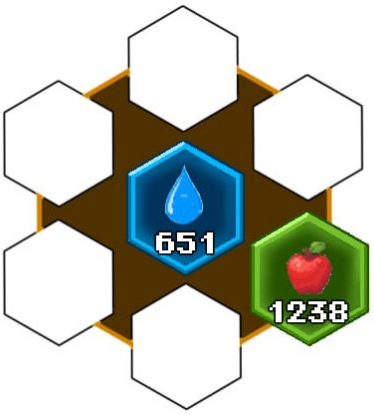
Minecraft - Main Menu

- **Gameplay Frame UI:** The external border housing colony-wide information, and gameplay icons to build.
- **Bee Interaction Radial:** Radial that shows all possible tasks a bee can perform when interacting with it.
- **Task Icons:** Show up in the bee's queue of tasks.



The Sims 2- Game UI with events, needs, construction

- **Hexagonal Resource Icons:** Individual hexagonal representations for Nectar, Pollen, Resin, Water, Glue, Food, and Honey.



Example of resources

- **Timeline Interface**: A horizontal bar showing past events (left), upcoming events (right), and the centered present moment, including current season and weather.
- **Event Icons:** Both for events in the timeline, and foraging events



Frostpunk - UI with seasons, weather

- **Time Controls**: Icons for Play/Pause and speed multipliers (1x, 2x, 3x).



City Skylines - Time controls

- **Mood Speech Bubbles**: Small visual indicators that appear above a bee's head to communicate negative states like "Angry" or "Tired".



Prison Architect - Intent Icons

- **Trait Icons:** Specific symbols for Life Span, Hunger, Speed, Size, Energy, Aggression, Sweetness, and Luck.
- **Event Interface:** A specific interface for events that occur to foraging bees, having a description of the event, options on how to react to the events, and outcomes of the events



Mewgenics - Events

- **Role Specialization Icons:** Distinctive hats to indicate assigned roles (Soldier, Worker, Foraging).

8.2 Environment Assets

The game world uses a lateral-cutaway perspective, revealing the interior of a tree trunk sectioned into hexagonal units.

- **Hive Tileset:** A minimal set of 14 hexagonal tiles (flat-top) that interconnect to form rooms, 4 variations of these 14 tiles for all 4 players.
- **Background Textures:** Deep brown textures for the empty trunk space and golden hues for established cells.

- **Workstations & Furniture:**
  - **Production Units**: Large furnaces to smelt nectar into honey, nurseries to let larvae grow into bees, storage units, food production units.
  - **Customization**: Furniture pieces stylized to fit within the hive. Couches, lamps, beds, wardrobes, lights, shelves, posters and much more.



Terraria - Building UI

- **World Borders:** Faded images of the outside world visible at the horizontal limits of the tree trunk width. Depending on seasons



- **Weather & Seasonal Overlays**: Visual effects for Summer (intense light/heat waves), Autumn (rain/flooding), and Winter (snow/greyish-blue tints).
- **Events**: Splash art for all the foraging events, in accordance to their flavor text.

8.3 Characters and Animation

All bee characters are designed with modularity in mind, using three distinct sprites (head, body, stinger) rigged to a shared armature to create an intentional "rag doll" movement.

- **Avatar**: The Queen bee, identified by a unique regal hat. 4 Different sprites for each multiplayer player.
- **Bee Species Sprites**: Individual assets for each body part of Honey, Leafcutter, Sweat, Cuckoo, Bumble, Killer, Carpenter, Japanese, Vulture, and Digger bees.



Bombus - Bumble Bee, with sprite example



Japanese, Leafcutter, Sweat, Vulture, Killer bees

- **Larvae**: Sprites for the incubation and growth phases within nurseries.
- **Enemy Insects**:
    - Giant Hornet: A boss-like enemy with a large health pool.
    - Wax Moth: A sneaky, camouflaged invader.



Giant Asian Hornet and Wax Moth

- **Mutations:** Sprites needed for all 9 mutations.

The following animations will be performed through bone rigging:

- **Animation List (Bees Shared Armature)**:
    - Default idle: Gentle sway of the head and stinger.
    - Foraging/Expedition: Bees departing the hive and returning with resources.
    - Combat: Swarming intruders of the hive. Jittering sprites as they surround the hornet
- **Hornet and Moth movements:** Swaying of their heads and bodies.

The following animations will be performed through custom sprite animations:

- **Movement:** Animations needed for all body pieces, wing buzzing.
- **Building/Working:** Specialized interactions with workstations or hexagonal tiles.
- **Resting:** Animations needed for all body and head pieces, to show a sleeping bee.
- **Death:** Final animation for bees that die within the hive.
- **Enemy combat:** Hornet destroying the nurseries with their stingers.



Don't Starve - Bee farm ALL ABOUT THE HONEY

Don't Starve here showcases animations that use a mixture of sprite animations for the buzzing of their wings, and bone rigging for the sway of their stigers.

- **Player Cursors:** color coded for each player, one for general play, and one hammer cursor for building mode

8.4 Music and Sound Effects (SFX)

The audio must enhance the contrast between the surreal warmth of the hive and the perilous natural world outside. Additionally, for better immersion in the game world, given the nature of the game, classical instruments are very effective at conveying a natural and lively atmosphere (such as piano, violin, viola...).

- **Music Tracks:**
  - Main Menu Theme consists of an arrangement of vocals, which don't sing worlds but Onomatopoeic sounds that evoke bees in harmony (and a soft base by piano).
  - Hive Management Theme: Surreal, calm, and cozy background music (i.e. Main sound track, Winter sound track, Autumn sound track, Spring sound track).
  - Invasion Theme: High-tension music for hornet attacks or moth infiltrations (i.e. Assault Sound track).
  - Seasonal Variants: Subtle tonal shifts in music based on the current season.
- **Sound Effects (SFX):**
  - Ambient Buzzing: Background drone that changes based on colony size and mood (i.e. Nature sounds).
  - Harvesting/Gathering: Feedback sounds for nectar, pollen, resin, and water collection.
  - Production: Sounds of giant furnaces and construction work.
  - Environmental SFX: Wind rustling branches, rain during autumn floods, and winter storms (i.e. Wind blowing).
  - UI Audio: Hexagonal button clicks and timeline event notifications.

8.5 Art style references







Specifically the 16-bit 2D artstyle of the game that precedes the 3D.

8.6 Fonts

Baloo 2:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut ut pulvinar leo, sit amet pulvinar tellus. Pellentesque facilisis dui sed metus.

Fuzzy Bubbles:
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut ut pulvinar leo, sit amet pulvinar tellus. Pellentesque facilisis dui sed metus.

Pixelify Sans:
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut ut pulvinar leo, sit amet pulvinar tellus. Pellentesque facilisis dui sed metus.

Other font options:

- M5x7 (by Daniel Lissen)
- Thaleah Fat
- Compass
- Cubic

9. Prototype Summary

Hivemind prototype is intended as a proof of concept, focusing on the most important pillars of our game:

1. **Listen server architecture**
2. **Save and load system**
3. **Concurrent hive editing system**
4. **Bees AI and breeding system**

This prototype would allow us to demonstrate the stability of listen server architecture by enabling testing of the simulation with up to the intended 4 players per session and testing the two main mechanics that need to be designed properly to handle conflict resolution.

9.1 Listen server architecture

With this feature we intend on implementing:

- **Host/Client initialization**:
  The player is supposed to be able to enter one of his already created hives while having the role of the host in single player mode.

- **Relay and lobby integration**:
  The host should have an available relay allocation obtained through Unity Relay Service and a room code given by Unity Lobby Service when entering a game in single player mode.

- **Session joining**:
  The host must be able to send a request to another player using the Unity Lobby join code and the host must be able to decide whether to accept or reject the request.

9.2 Save and load system

With this feature we intend on implementing:

- **Hive creation**:
  The player must be able to create a new hive based on a standard template and start a play session with it.

- **Hive data save**:
  The game must be able to save the progress made while playing inside a hive. Hive data must be initially saved when creating a hive and later when edited, locally inside the player machine with a json file.
  The game additionally saves the summary of hives created by the player to display in the hive selection screen.

- **Hive data load**:
  The game must be able to retrieve the data relative to a hive and load it to display it in the same way it appeared before ending a play session with the same hive.
  The game must also be able to load the summary and display it in the hive selection screen.

9.3 Concurrent hive editing system

With this feature we intend on implementing:

- **Cell edit mode**:
  Multiple players must need to be able to edit the merged hive concurrently.
  Players must be able to choose a building target and add cells to their hive.
  Players must be able to delete cells from the merged hive, while still satisfying the connectivity checks.
  Players in edit mode must be able to see the modifications being made live by other players, while players not in edit mode must be able to see the modifications only once they have been

committed.
All pending additions and deletions must be synchronized between all players in cell edit mode, so that to avoid or easily resolve conflicts while trying to interact with the same cell at the same time.

- **Furniture edit mode**:
  Players must be able to place different furniture inside the hive, while respecting placement rules. Furniture can also be rotated 60° or mirrored before being placed.

- **Workbench edit mode**:
  Players must be able to place different workbenches inside the hive, while respecting placement rules.

9.4 Bees AI and breeding system

With this feature we intend on implementing:

- **AI behaviour**:
  The basic underlying AI piloting the bees should be influenced by the bee's trait values.

- **Breeding**:
  The implementation will be simplified and not fully representative of the original concept.
  The queen tries to deposit a larva every 6 minutes, checking first for available space in any nursery. If the hive meets the required conditions (food level, bee population, consumption, and production rates), she will lay an egg and restart the cycle. The decision process is intentionally probabilistic and not perfectly optimal, allowing for natural variation in colony behavior. Once an egg is placed, it becomes a potential target for genetic interaction from other bees over time.